

ELEC6910J

Deep Reinforcement Learning

Lecture 1. Introduction & Probability Basics

Course Instructor: Ling PAN

Department of Electronic and Computer Engineering

Hong Kong University of Science and Technology

Teaching Team

- Instructor: Ling PAN (lingpan@ust.hk)
Office hour: 18:00-19:00, Friday
- Teaching assistants:



Yuxiao YE

yyebs@connect.ust.hk

Office hour: 7-8pm, Wednesday

[meeting access omitted]



Ziqi WANG

zwangnp@connect.ust.hk

Office hour: 8-9pm, Wednesday

[Zoom](#):

- [meeting access omitted]

- [meeting access omitted]

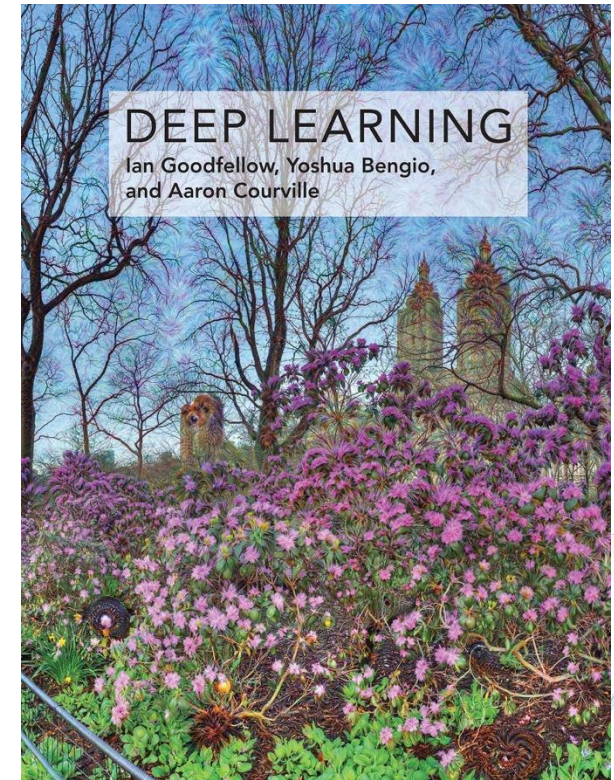
Feel free to let us know if you have
course related questions



Logistics

*Deep Reinforcement Learning =
Deep Learning + Reinforcement Learning*

- Slides will be uploaded to Canvas after lectures
- Required background:
 - Deep learning
 - Probability basics (will be reviewed in this lecture)
 - Python/PyTorch/... programming
- Reference book (not required):
 - Reinforcement learning, Barto and Sutton 1998.



Reinforcement Learning

An Introduction
second edition

Richard S. Sutton and Andrew G. Barto

Logistics

- Honor Code
 - You agree not to perform any action that may cause plagiarism
 - Will lead an F for this course
 - Zero tolerance

Schedule

- Lecture videos will be posted on Canvas 2 weeks before the exam
- Important dates:
 - Mid-term exam: **Oct. 23**
 - Project presentation: **Nov. 20, Nov. 27**

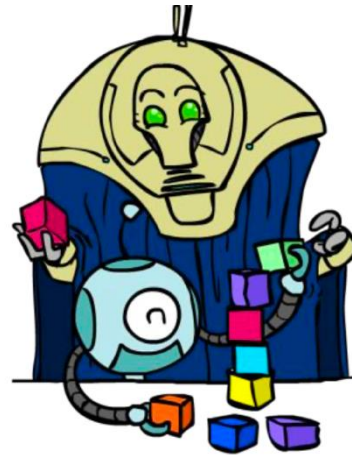
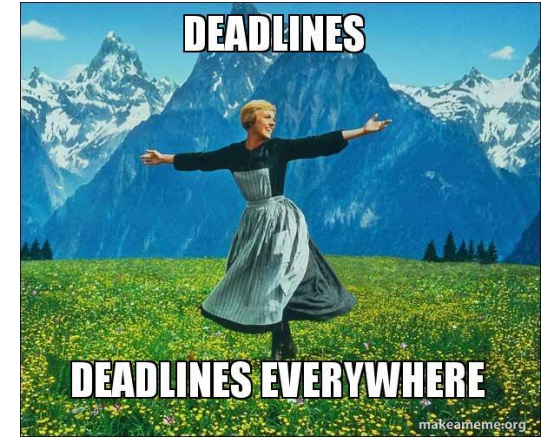
Project Presentation	Mid-term Exam	Lecture	Holiday				
Wk	September						
	Sun	Mon	Tue	Wed	Thu	Fri	Sat
1			1	2	3	4	5
2	6	7	8	9	10	11	12
3	13	14	15	16	17	18	19
4	20	21	22	23	24	25	26
5	27	27	29	30			
	October						
	Sun	Mon	Tue	Wed	Thu	Fri	Sat
5					1	2	3
6	4	5	6	7	8	9	10
7	11	12	13	14	15	16	17
8	18	19	20	21	22	23	24
9	25	26	27	28	29	30	31
	November						
	Sun	Mon	Tue	Wed	Thu	Fri	Sat
10	1	2	3	4	5	6	7
11	8	9	10	11	12	13	14
12	15	16	17	18	19	20	21
13	22	23	24	25	26	27	28

Waiting List

- Current Quota: 30 students
 - After thorough discussion, we are unable to increase the quota at this time.
 - We need to limit the size to our current capacity to ensure every enrolled student receives the necessary attention and guidance to excel in RL.
- For Waiting list students:
 - If you are highly interested, please remain on the waiting list.
 - Spots may become available if currently enrolled students decide to drop the course.

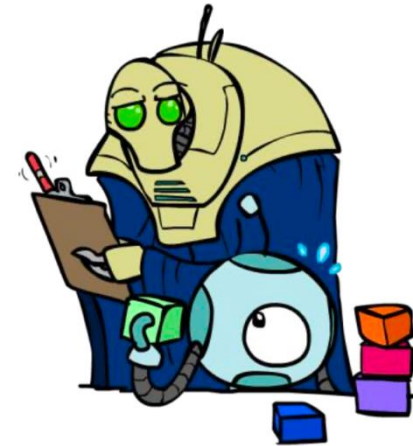
Grading

- Attendance (10%)
- Assignments (20%)
 - Your homework will NOT be accepted if you are late.
 - Everyone has 3 free late days in total for this whole semester. TAs will track this.
- Exam (10%)
 - IMPORTANT: October 23
- Project
 - Presentation: 30%
 - Report: 30%



Instruction

Grow knowledge,
collaborate,
work until success



Assessment

Measure knowledge,
each student on their own,
stopped before success

Topics

- Introduction and Probability Basics
- Supervised Learning Basics,
- Bandits, Exploration, MDP
- Dynamic Programming, Monte Carlo Methods
- Temporal Difference Learning
- Policy Gradients, Actor-Critic Methods
- Advanced Applications of DRL

Outline

- Introduction
- Probability Basics

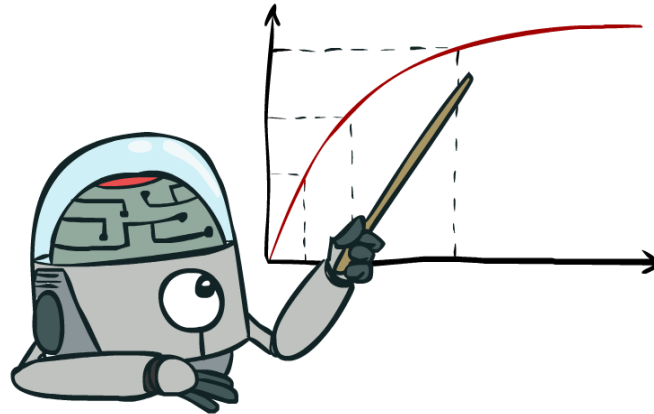
What is AI?

The science of making machines that:



Rational Decisions

Maximize Your Expected Utility



Machine Learning

- Arthur Samuel (1959): Machine Learning is the field of study that gives the computer the ability to learn without being explicitly programmed.
- Tom Mitchell (1998): A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .



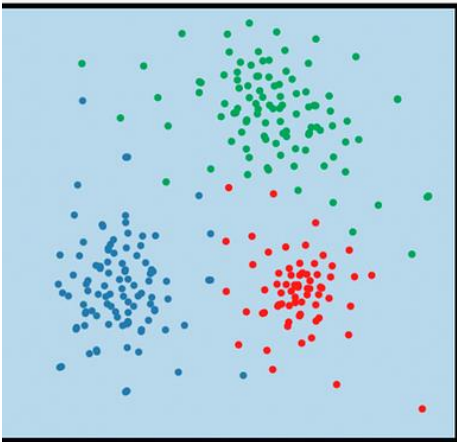
PEAS

- Use PEAS to describe task
 - Performance measure
 - Environment
 - Actuators
 - Sensors
- Example: Taxi driver
 - Performance measure: safe, fast, comfortable (maximize profits)
 - Environment: roads, other traffic, pedestrians, customers
 - Actuators: steering, accelerator, brake, signal, horn
 - Sensors: cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors
- Environment properties
 - Fully observable vs. partially observable
 - Deterministic vs. stochastic
 - Static vs. dynamic
 - Discrete vs. continuous
 - Single agent vs. multiagent

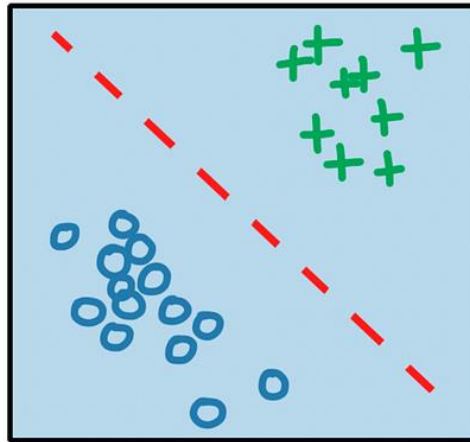
Machine Learning

machine learning

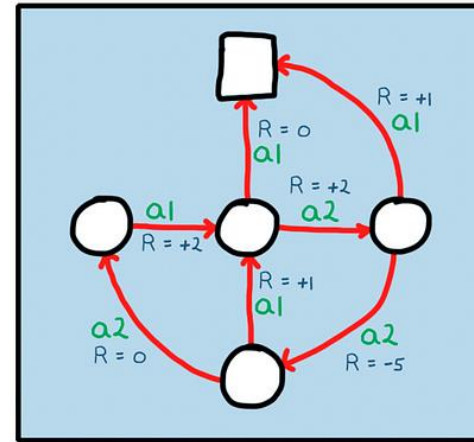
unsupervised
learning



supervised
learning

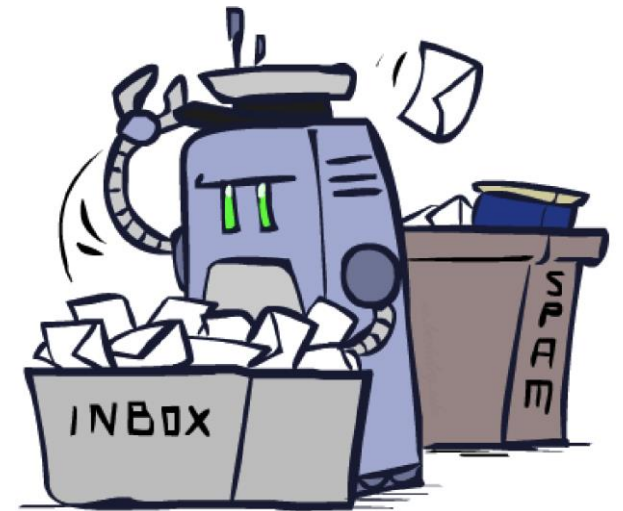


reinforcement
learning



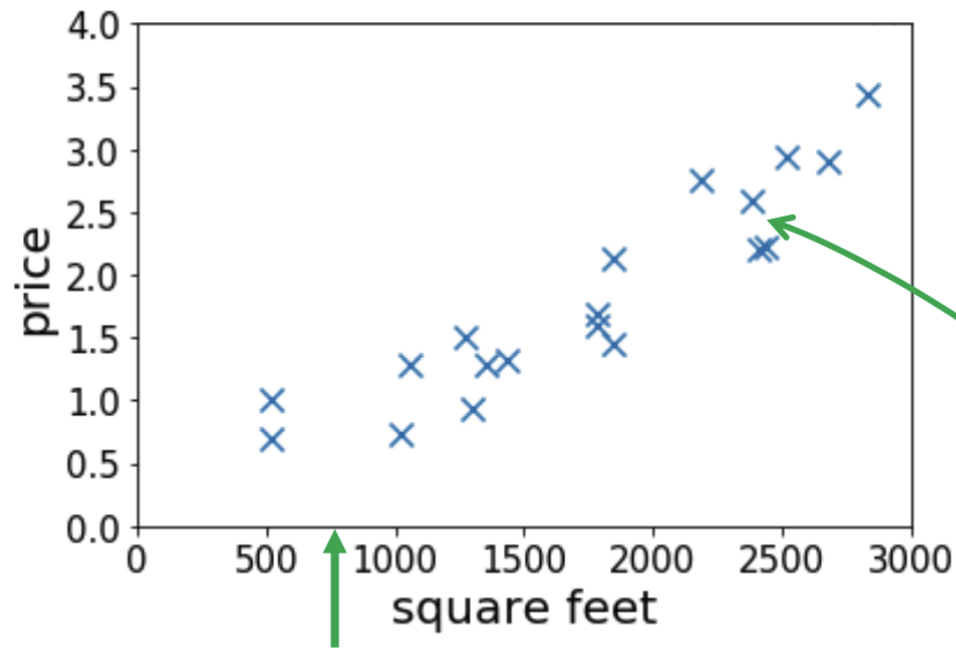
Supervised Learning

- Regression: predicts a numeric value
 - Example: future house price prediction, ride time estimation, ...
- Classification: predicts if something belongs to a particular category, and it could be binary or multiclass classification
 - Example: spammed emails, ...

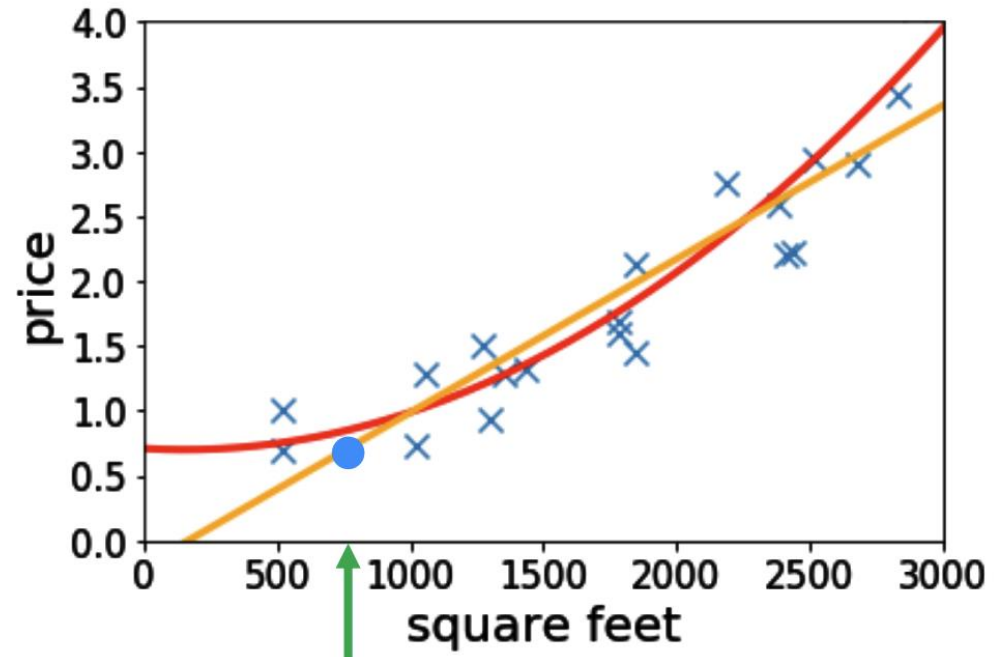


Supervised Learning: Regression

- Given: a dataset that contains n samples $((x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}))$
- Task: if a residence has x square feet, predict its price?



$x = 800$
 $y = ?$



$x = 800$
 $y = ?$

Supervised Learning: Classification

- Example: Spam filter
 - Input: an email
 - Output: spam/ham
- Setup:
 - Get a large collection of example emails, each labeled “spam” or “ham”
 - Note: someone has to hand label all this data!
 - Want to learn to predict labels of new, future emails
- Features: The attributes used to make the ham/spam decision
 - Words: FREE!
 - Text Patterns: \$dd, CAPS
 - Non-text: SenderInContacts, WidelyBroadcast
 - ...

Dear Sir.

First, I must solicit your confidence in this transaction, this is by virtue of its nature as being utterly confidential and top secret. ...

TO BE REMOVED FROM FUTURE MAILINGS, SIMPLY REPLY TO THIS MESSAGE AND PUT "REMOVE" IN THE SUBJECT.

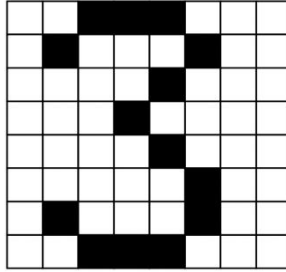
99 MILLION EMAIL ADDRESSES
FOR ONLY \$99

Ok, I know this is blatantly OT but I'm beginning to go insane. Had an old Dell Dimension XPS sitting in the corner and decided to put it to use, I know it was working pre being stuck in the corner, but when I plugged it in, hit the power nothing happened.

Supervised Learning: Classification

- Example: Digit recognition

- Input: images
- Output: a digit 0-9

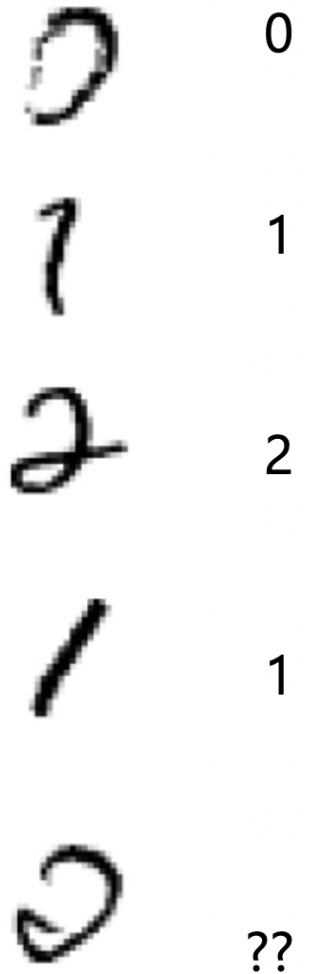


- Setup:

- Get a large collection of example images, each labeled with a digit
- Note: someone has to hand label all this data!
- Want to learn to predict labels of new, future digit images

- Features: The attributes used to make the digit prediction

- Pixels: (6,8)=ON
- Shape Patterns: NumComponents, AspectRatio, NumLoops
- ...



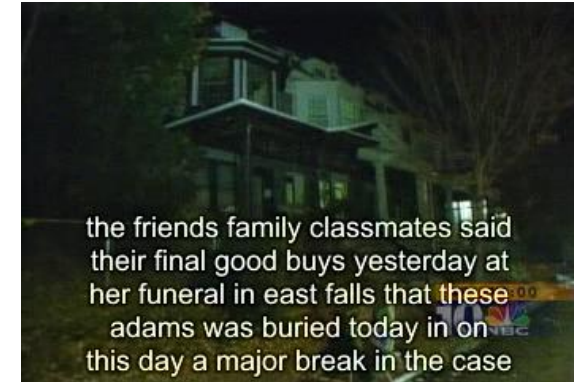
Natural Language Processing

- Fundamental goal: analyze and process human language, broadly, robustly, accurately...
- End systems that we want to build:
 - Modest: spelling correction, text categorization...
 - Ambitious: speech recognition, machine translation, information extraction, dialog interfaces, question answering...



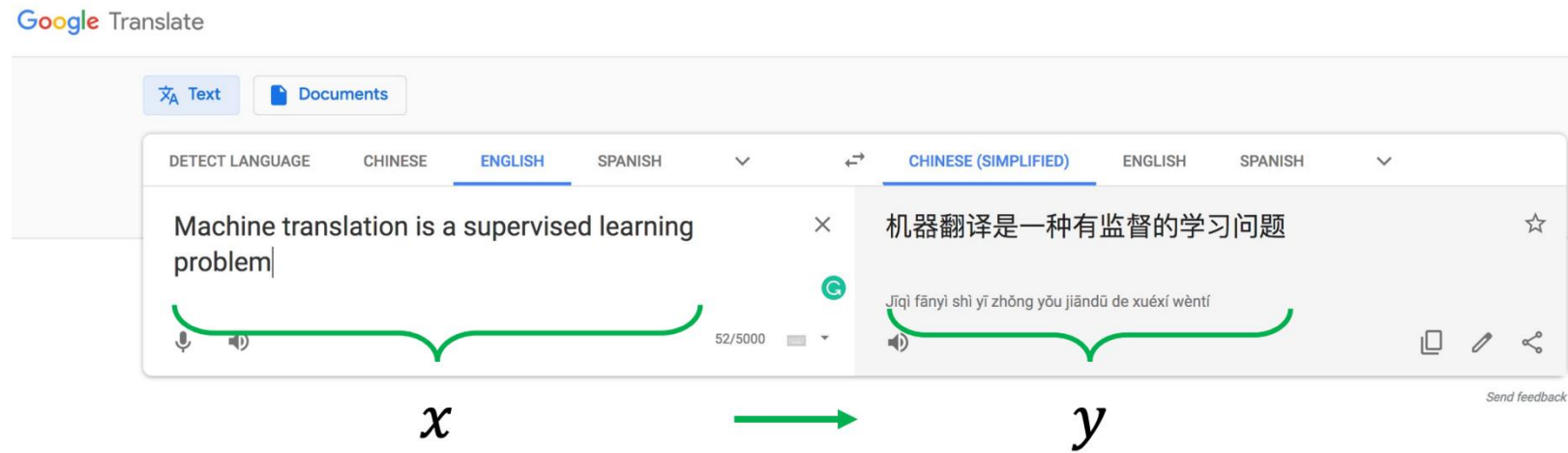
Natural Language Processing

- Speech technologies (e.g., Siri)
 - Automatic speech recognition (ASR)
 - Text-to-speech synthesis (TTS)
 - Dialog systems
- Language processing technologies
 - Question answering
 - Machine translation
 - Web search
 - Text classification, spam filtering, etc...



Supervised Learning in NLP

- Machine translation

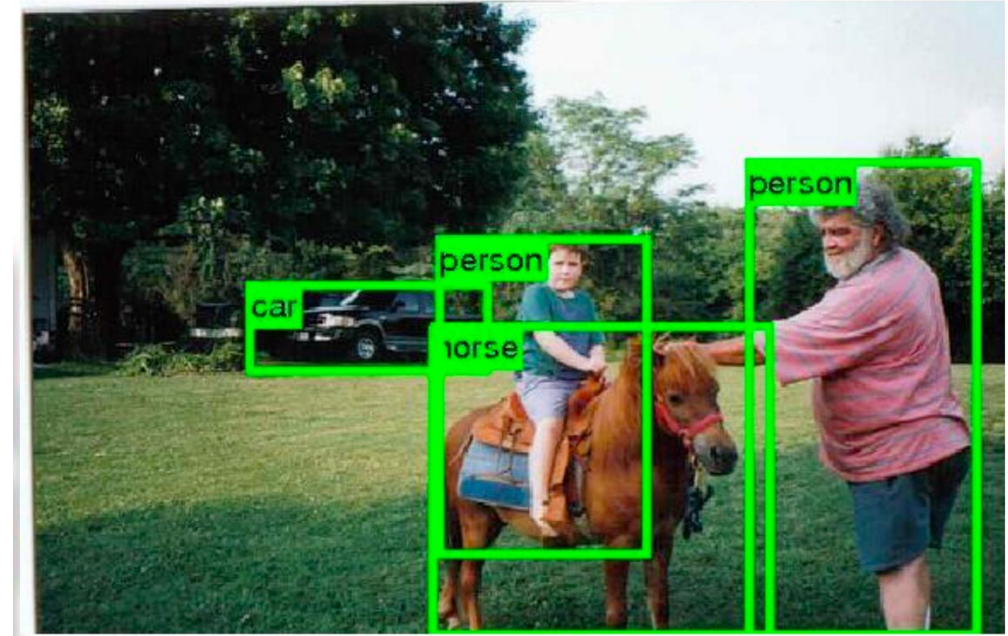


ChatGPT



Computer Vision

- Object and face recognition
- Scene segmentation
- Image classification
- ...



Supervised Learning in CV

- Image classification
 - x =raw pixels of the image, y =the main object



flamingo



cock



ruffed grouse



quail



partridge

...



Egyptian cat



Persian cat



Siamese cat

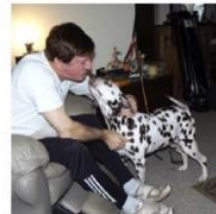


tabby



lynx

...



dalmatian



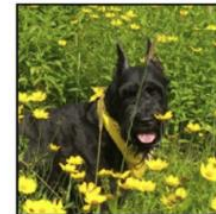
keeshond



miniature schnauzer



standard schnauzer

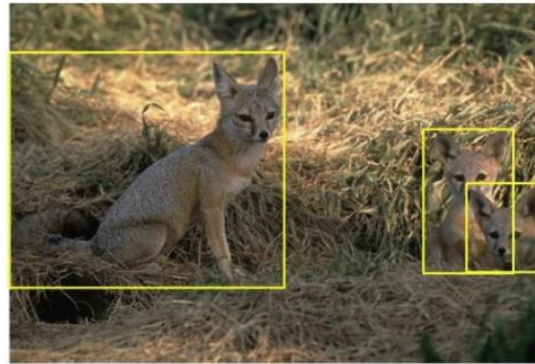


giant schnauzer

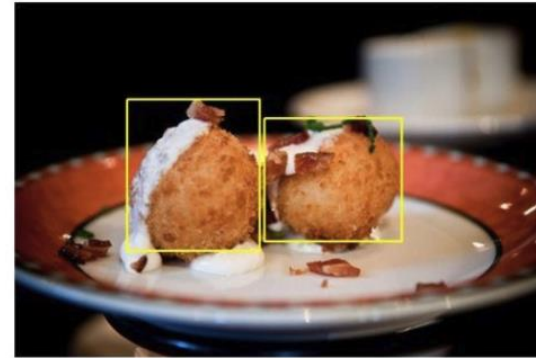
...

Supervised Learning in CV

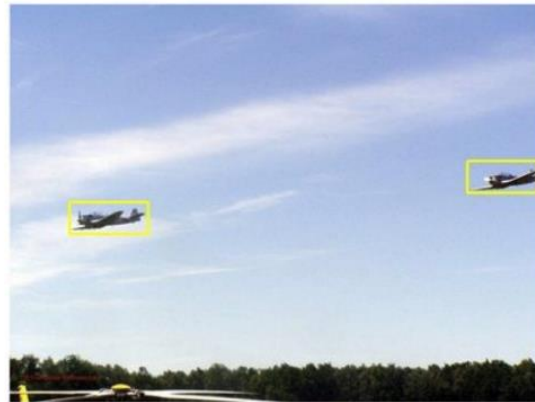
- Object localization and detection
 - x =raw pixels of the image, y =the bounding boxes



kit fox



croquette

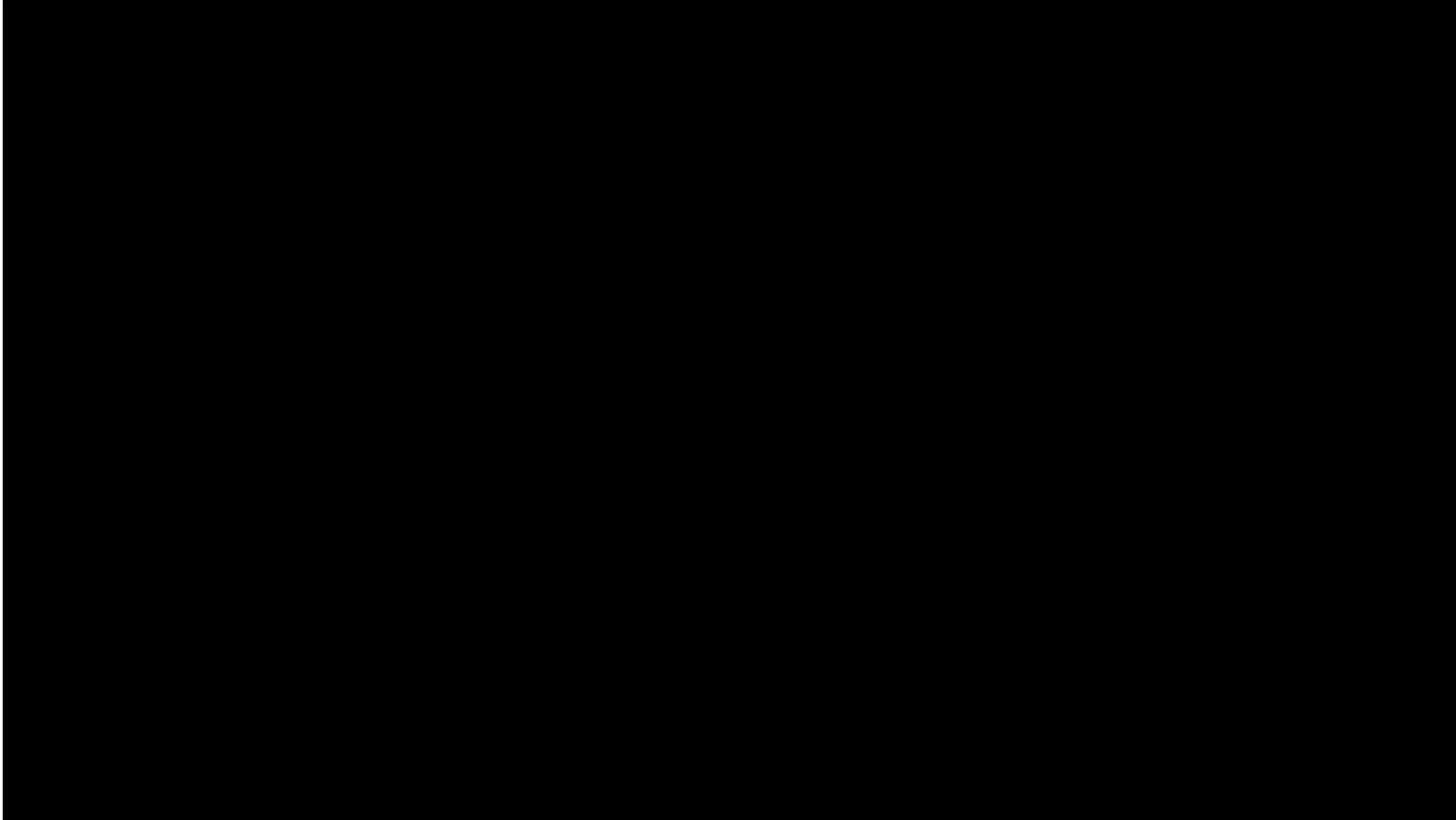


airplane

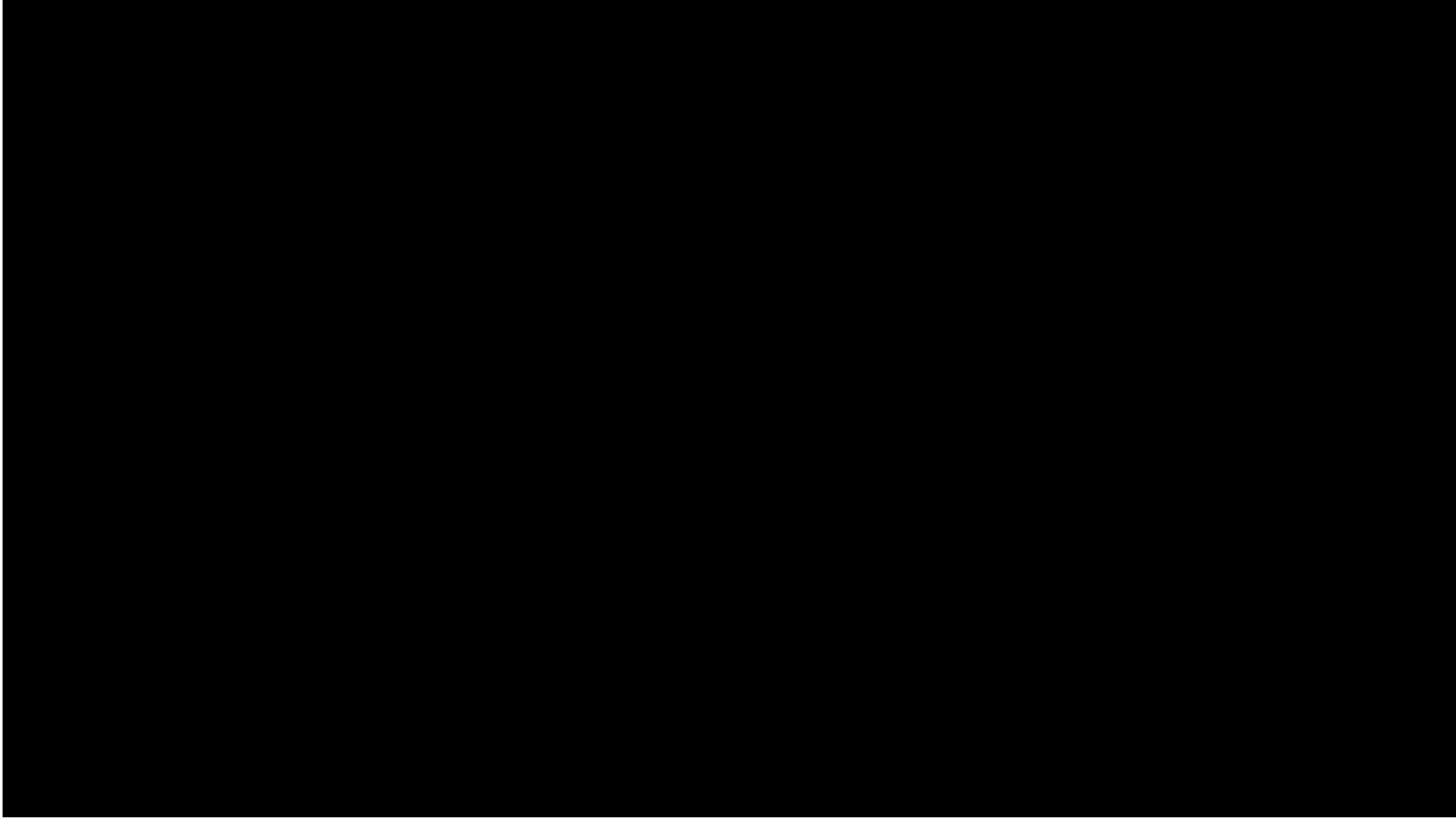


frog

Autonomous Driving



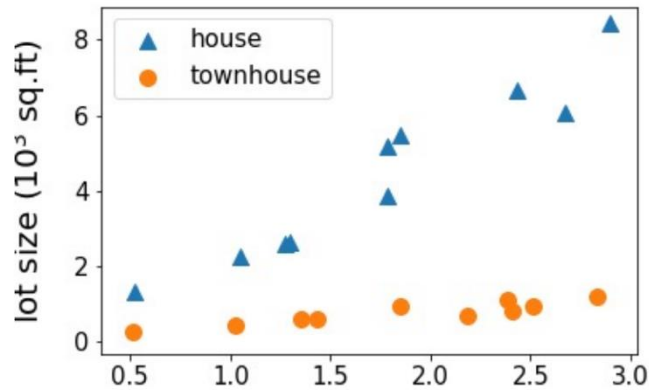
Stable Diffusion



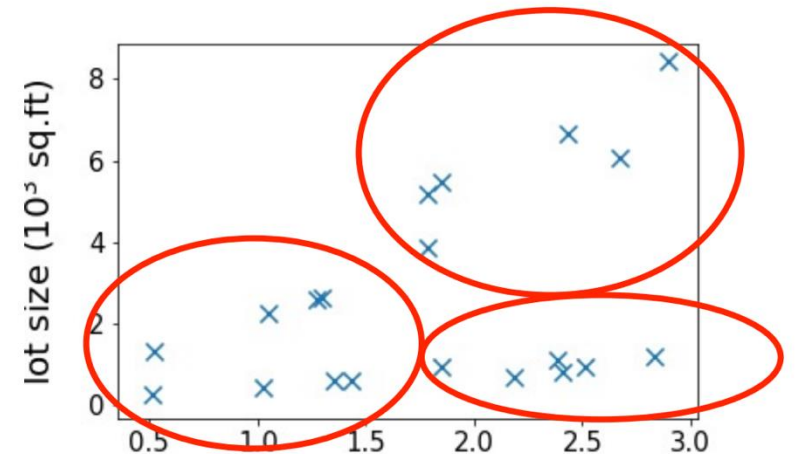
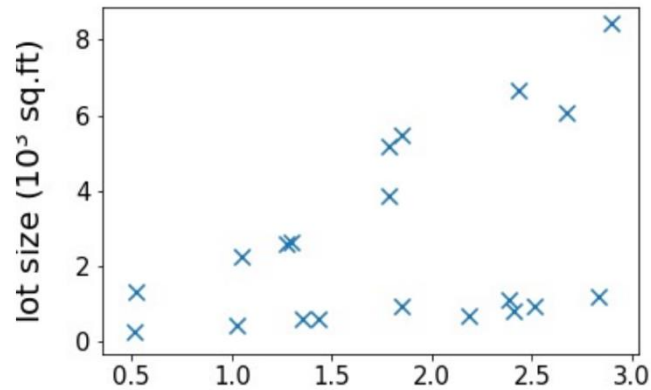
Unsupervised Learning

- Identify the hidden patterns in the data, e.g., clustering
- Dataset contains no labels: $x^{(1)}, \dots, x^{(n)}$
- Goal (vague): to find interesting structures in the data

supervised

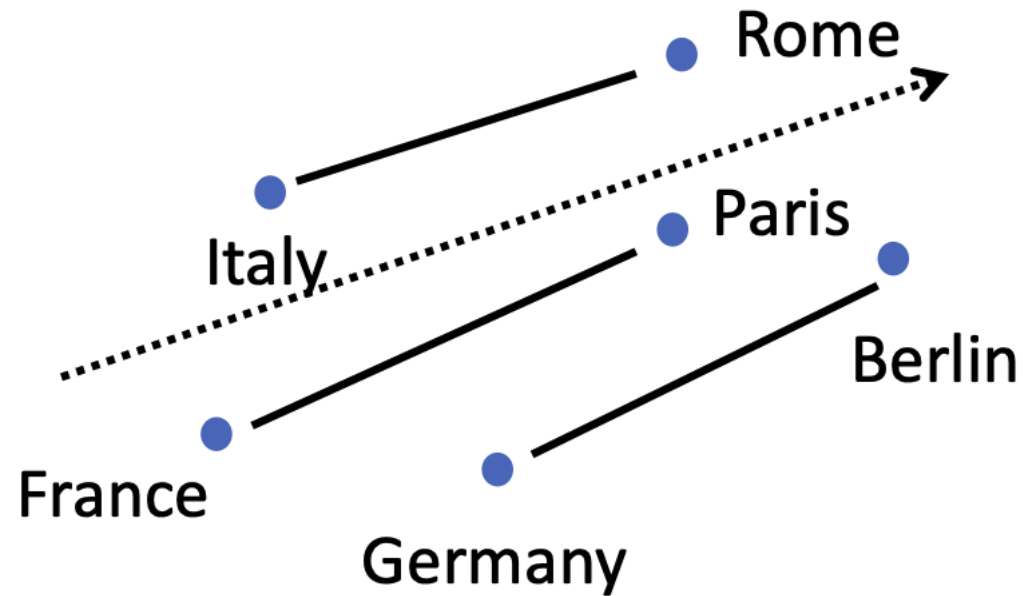


unsupervised



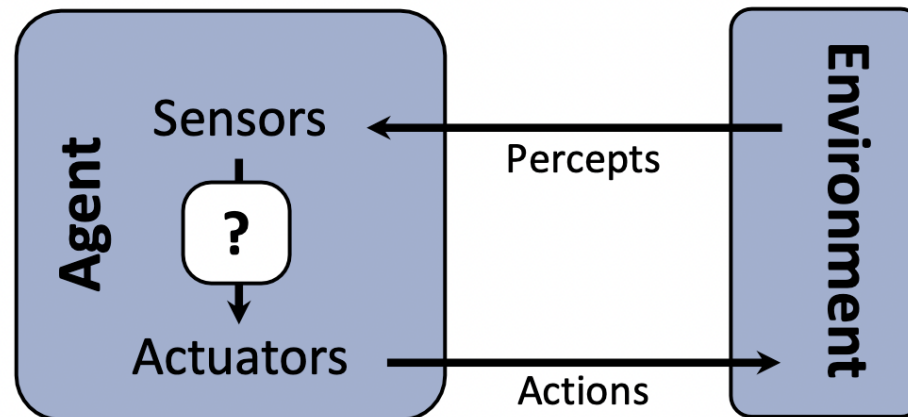
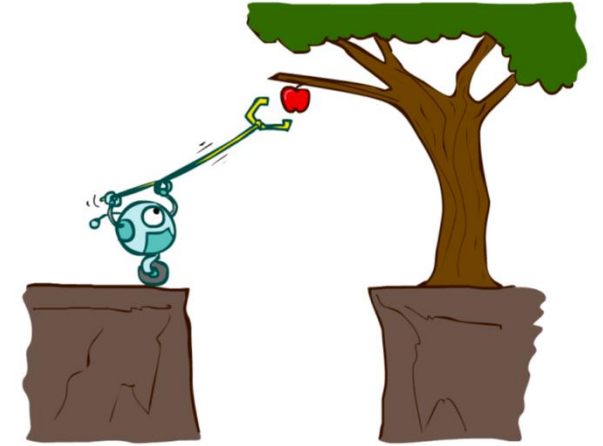
Unsupervised Learning

- Example: Word embeddings
 - Represent words by vectors
 - Represent relations by directions



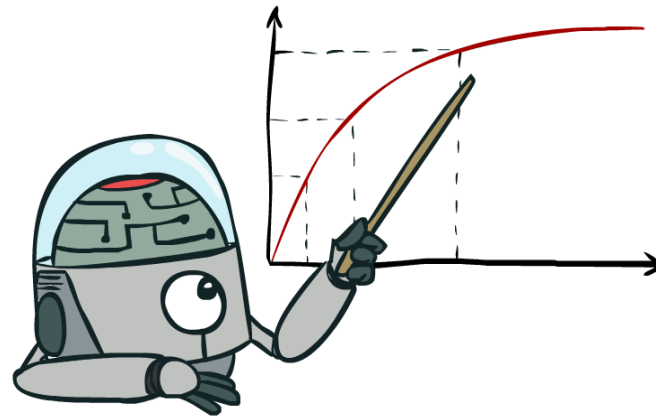
Designing Rational Agents

- An agent is an entity that perceives and acts
- A rational agent selects actions that maximize its (expected) utility
- Characteristics of the percepts, environment, and action space dictate techniques for selecting rational agents



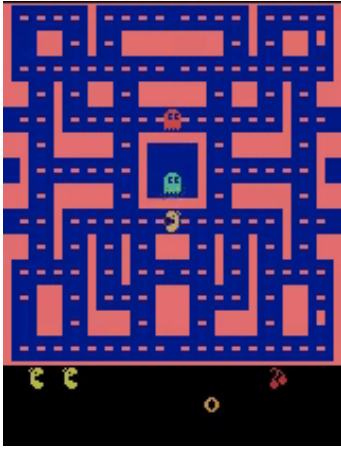
Rational Decisions

Maximize Your Expected Utility

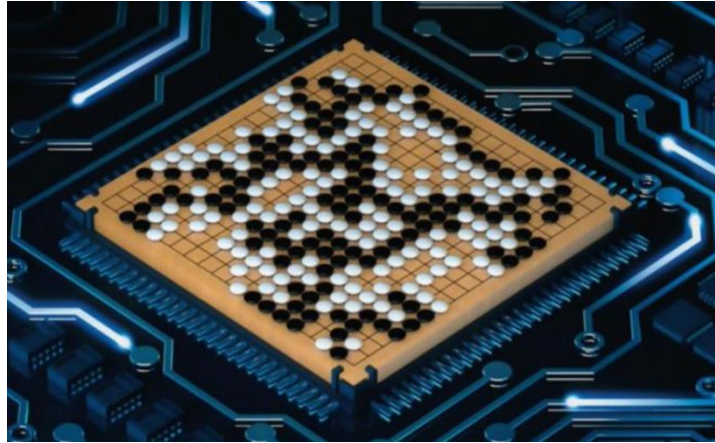


→ Reinforcement
Learning

Reinforcement Learning



Atari



Go



StarCraft II



Robotics



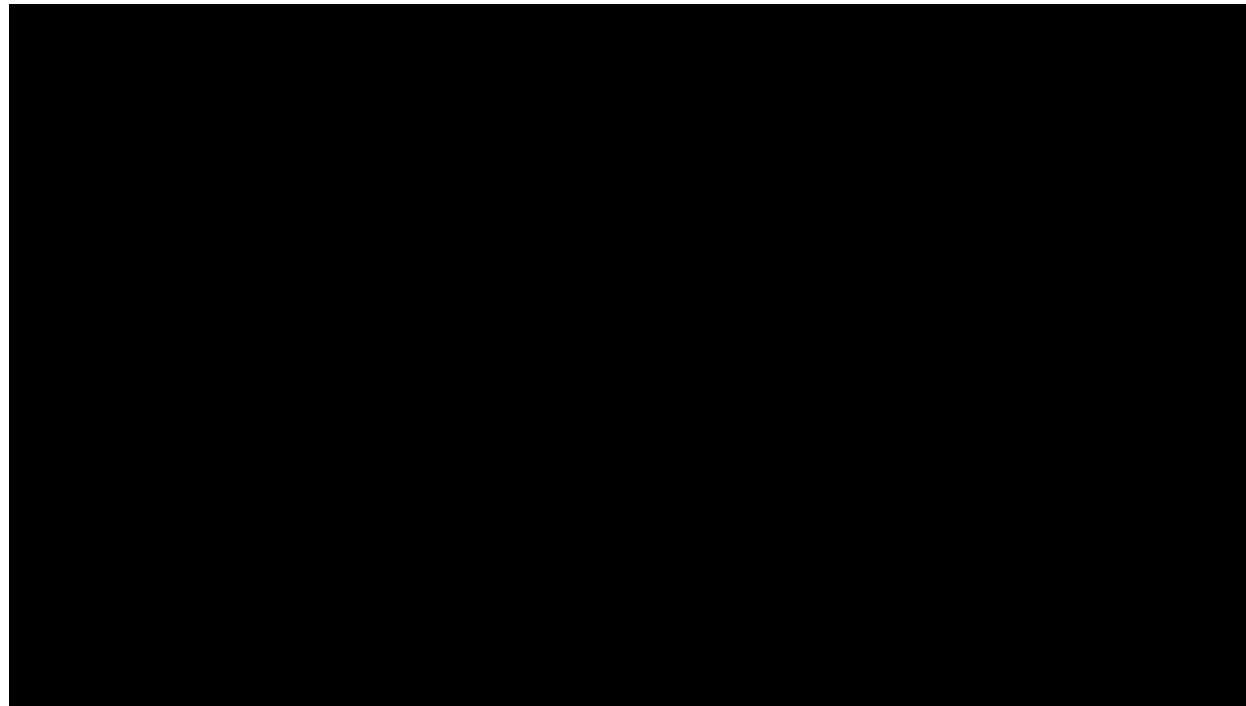
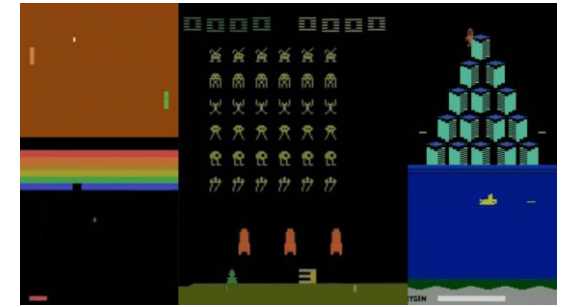
Transportation



Dialogue systems

Examples of DRL

- 2013: Playing Atari games



Starting out - 10 minutes of training

The algorithm tries to hit the ball back, but
it is yet too clumsy to manage.

Examples of DRL

- 2015: Locomotion from trial and error

DeepMimic: Example-Guided Deep Reinforcement
Learning of Physics-Based Character Skills
extra supplementary video



Xue Bin Peng¹, Pieter Abbeel¹, Sergey Levine¹, Michiel van de Panne²

¹ University of California
Berkeley 

² University of British
Columbia 

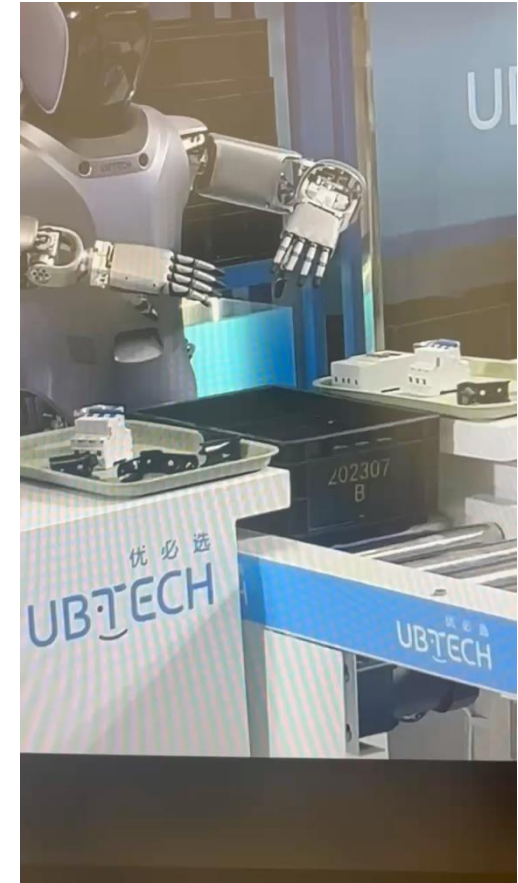
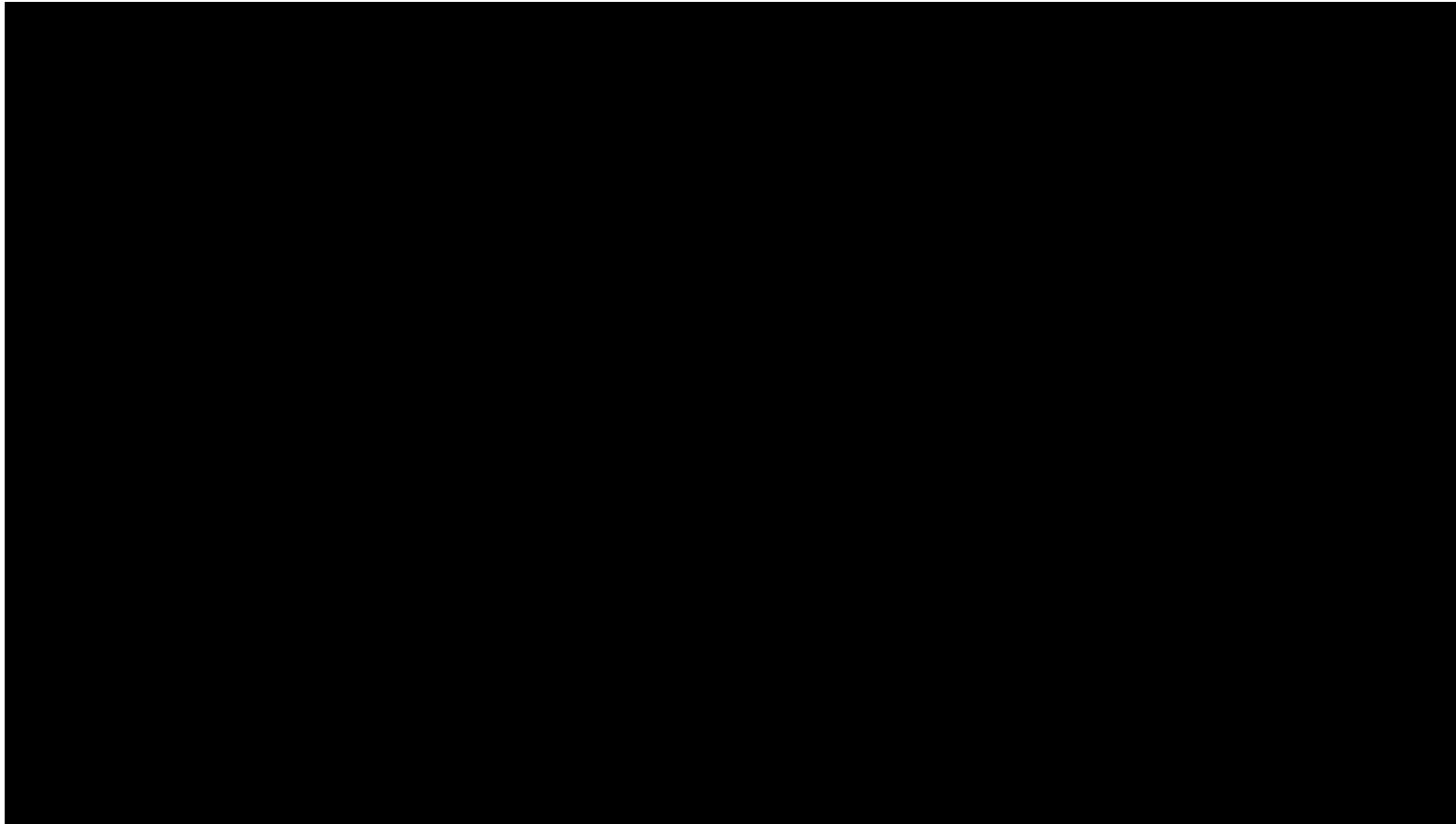
Examples of DRL

- 2016: Playing Go (and beating human champion)



Examples of DRL

- 2019: Robot manipulation



Examples of DRL

- 2019: Dota2
 - OpenAI beats OG world champions team of 5 pros
 - AI controlled 5 bots using different layers of same network
 - Trained with RL and self play
 - Equivalent to 45,000 years (over 10 months)



NoobFromUA

Examples of DRL

- 2022: Nuclear fusion plasma control

nature

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Article | [Open access](#) | Published: 16 February 2022

Magnetic control of tokamak plasmas through deep reinforcement learning

[Jonas Degraeve](#), [Federico Felici](#) , [Jonas Buchli](#) , [Michael Neunert](#), [Brendan Tracey](#) , [Francesco Carpanese](#), [Timo Ewalds](#), [Roland Hafner](#), [Abbas Abdolmaleki](#), [Diego de las Casas](#), [Craig Donner](#), [Leslie Fritz](#), [Cristian Galperti](#), [Andrea Huber](#), [James Keeling](#), [Maria Tsimpoukelli](#), [Jackie Kay](#), [Antoine Merle](#), [Jean-Marc Moret](#), [Seb Noury](#), [Federico Pesamosca](#), [David Pfau](#), [Olivier Sauter](#), [Cristian Sommariva](#), [Stefano Coda](#), [Basil Duval](#), [Ambrogio Fasoli](#), [Pushmeet Kohli](#), [Koray Kavukcuoglu](#), [Demis Hassabis](#) & [Martin Riedmiller](#) — [Show fewer authors](#)

[Nature](#) **602**, 414–419 (2022) | [Cite this article](#)

237k Accesses | **2419** Altmetric | [Metrics](#)

Demis Hassabis Facts



Demis Hassabis
Nobel Prize in Chemistry 2024

Born: 27 July 1976, London, United Kingdom

Affiliation at the time of the award: Google DeepMind,
London, United Kingdom

Prize motivation: “for protein structure prediction”

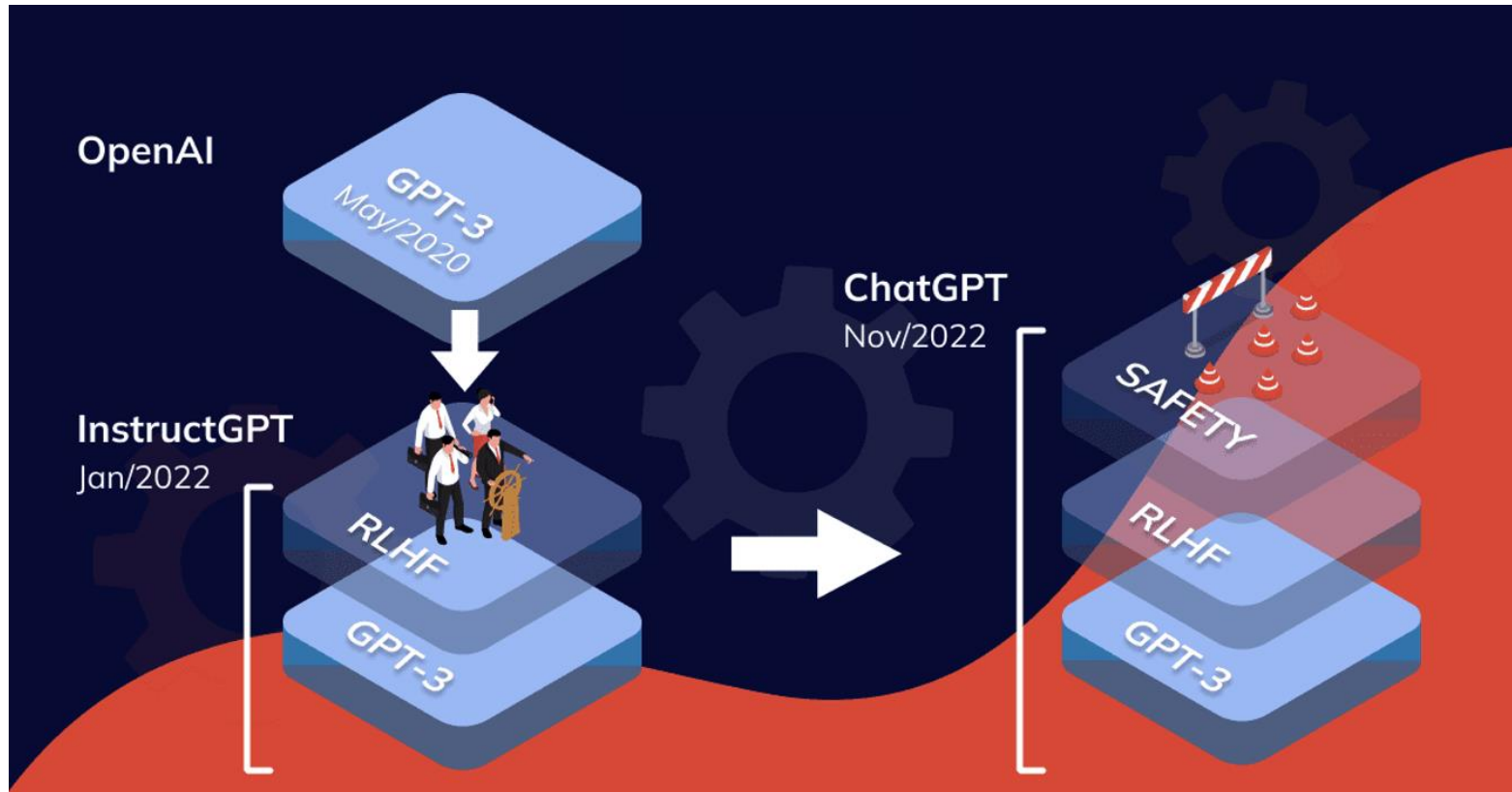
Prize share: 1/4

© Nobel Prize Outreach.
Photo: Clément Morin



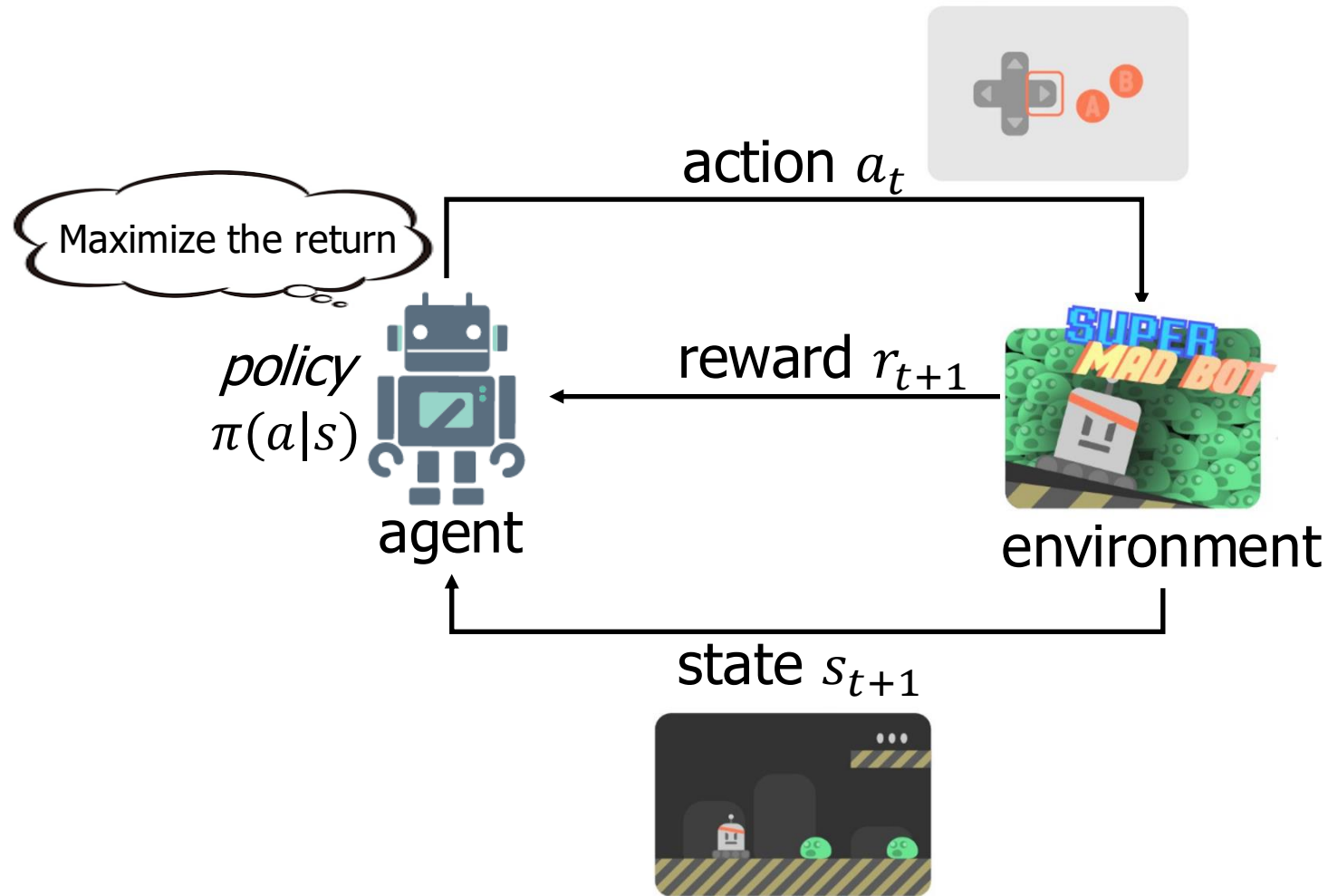
Examples of DRL

- 2022: Training language assistants with human feedback



- 2024: DeepSeek-R1 with Reinforcement Learning

Reinforcement Learning



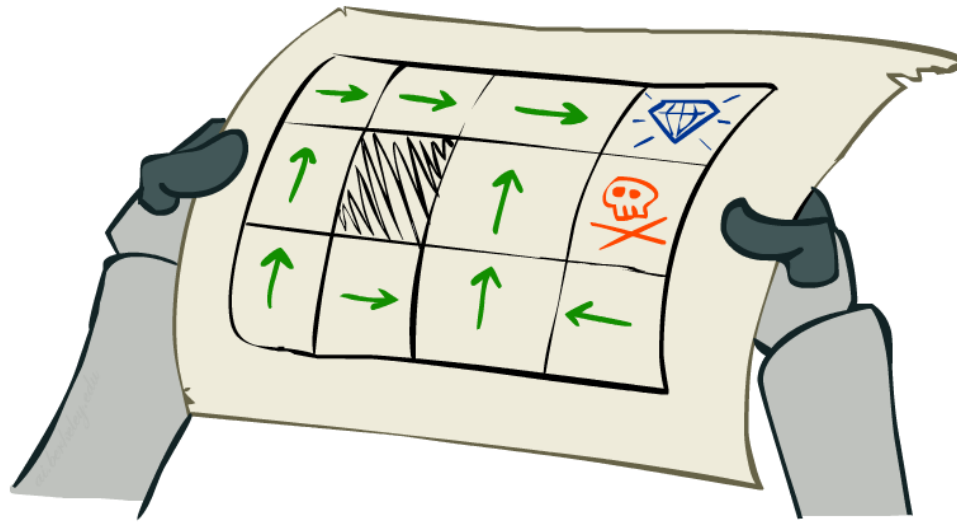
Model: Markov Decision Process (MDP)

- States $s \in S$
- Actions $a \in A$
- Transition dynamics $P(s'|s, a)$
- Reward function $R(s, a, s')$
- Discount factor γ

Goal: learn a policy π to maximize the expected long-term rewards

Reinforcement Learning

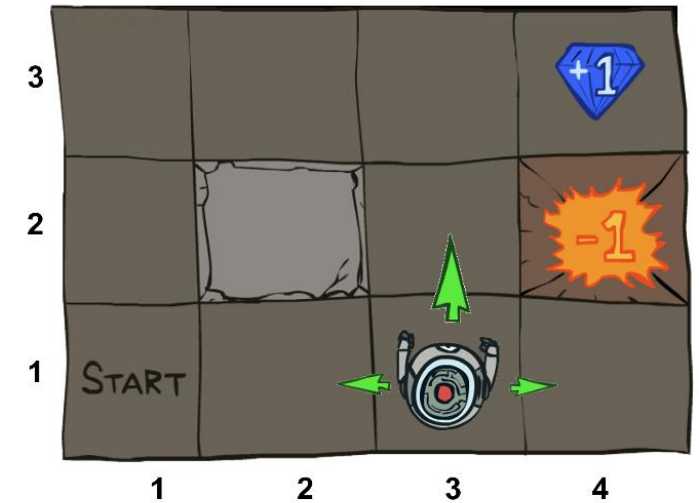
- In deterministic single-agent search problems, we wanted an optimal **plan**, or sequence of actions, from start to a goal
- For MDPs, we want an optimal **policy** $\pi^*: S \rightarrow A$
 - A policy π gives an action for each state
 - An optimal policy is one that maximizes expected utility if followed
 - An explicit policy defines a reflex agent



Optimal policy when $R(s, a, s') = -0.03$ for all non-terminals s

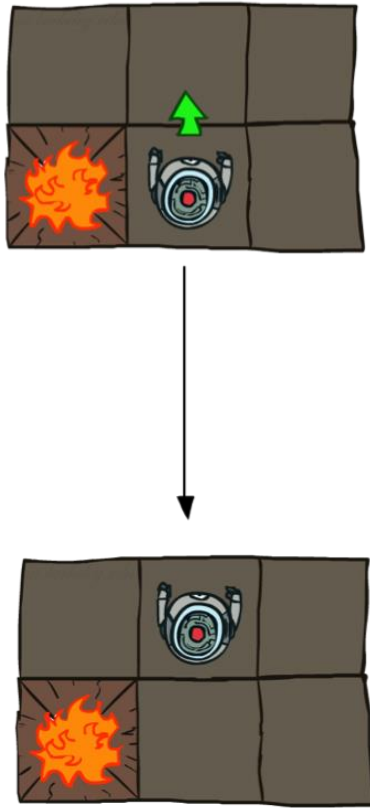
Reinforcement Learning

- A maze-like problem
 - The agent lives in a grid
 - Walls block the agent's path
- Noisy movement: actions do not always go as planned
 - 80% of the time, the action North takes the agent North (if there is no wall there)
 - 10% of the time, North takes the agent West; 10% East
 - If there is a wall in the direction the agent would have been taken, the agent stays put
- The agent receives rewards each time step
 - Small "living" reward each step (can be negative)
 - Big rewards come at the end (good or bad)
- Goal: maximize sum of rewards

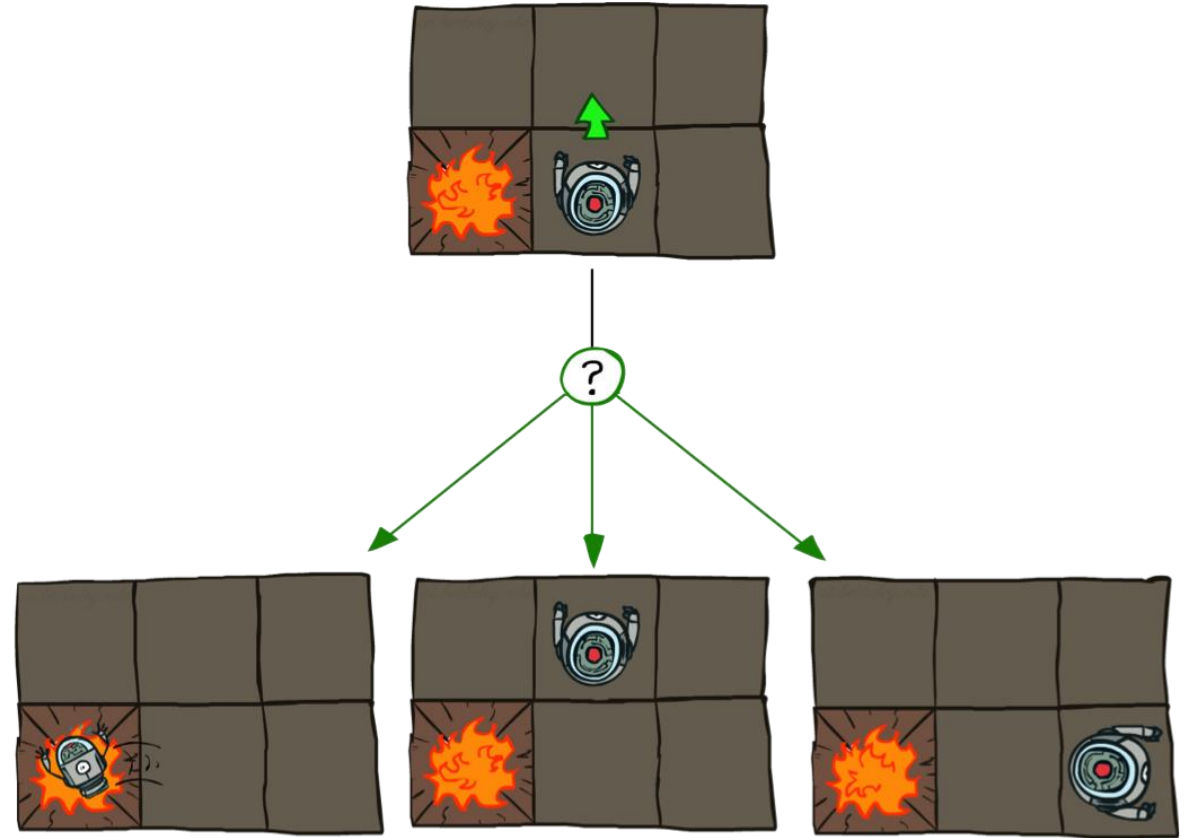


Reinforcement Learning

Deterministic Grid World



Stochastic Grid World



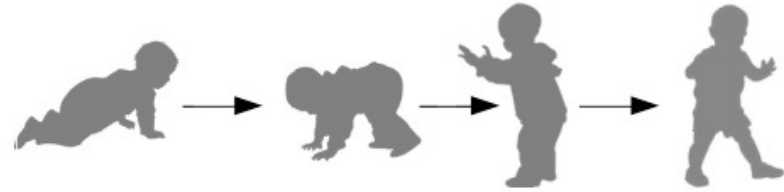
Reinforcement Learning

Problem: we don't know P or R



Which states/actions are good?

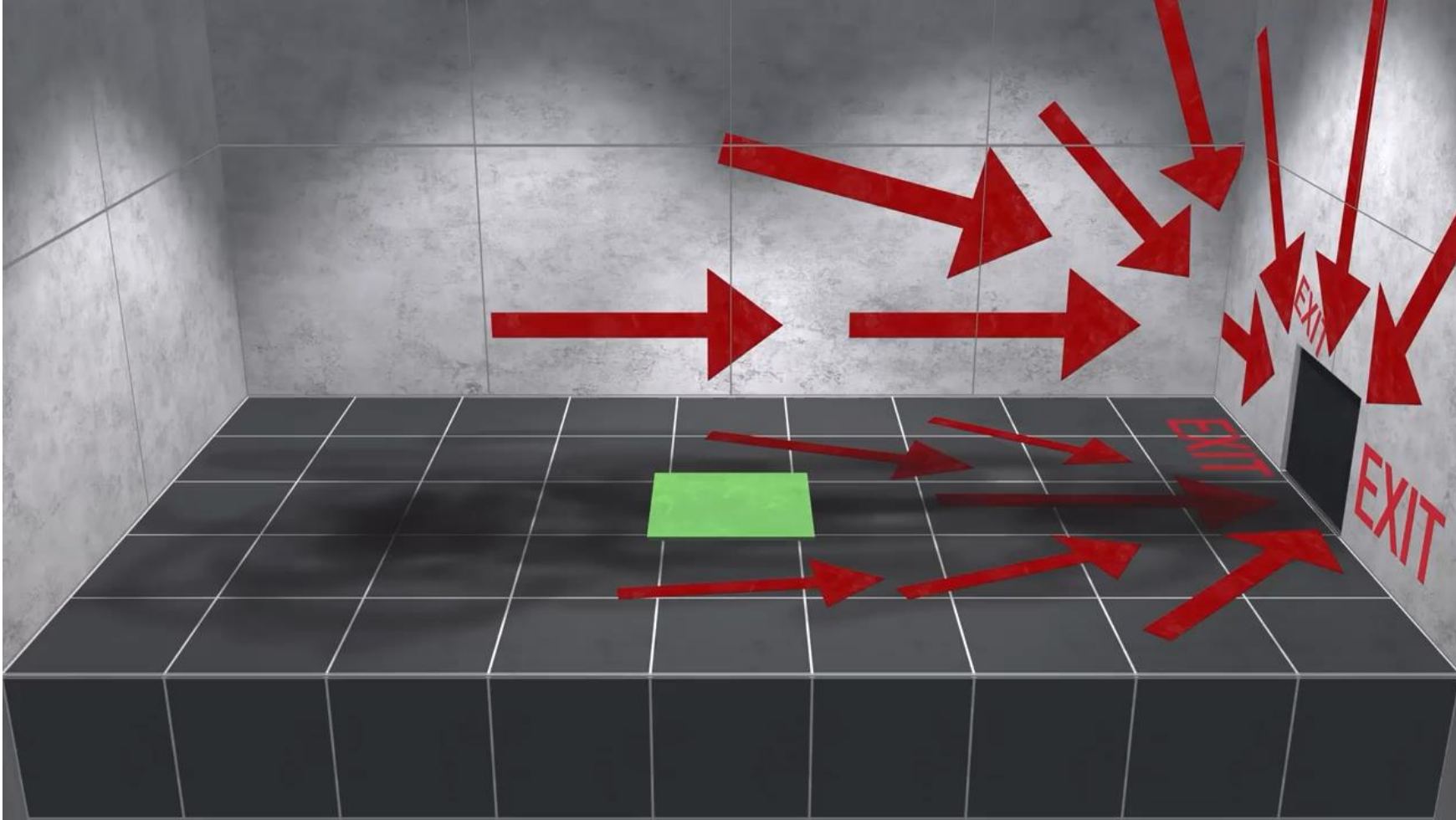
- Value function: $V^\pi(s)$ / $Q^\pi(s, a)$
→ Learn by **trial and error**



Expert Crawler
(7 months)



Deep Reinforcement Learning



Robotics



Setting up a Reinforcement Learning Task with a Real-World Robot

A. Rupam Mahmood, Dmytro Korenkevych,
Brent J. Komer, and James Bergstra



Robotics



Multi-Agent RL

- Hide and Seek



Multi-Agent
Hide and Seek

Multi-Agent RL

- StarCraft II



- Why is Starcraft hard?
 - The game of Starcraft is:
 - Adversarial
 - Long Horizon
 - Partially Observable
 - Realtime
 - Huge branching factor
 - Concurrent
 - Resource-rich
 - ...

ELEC6910J

Deep Reinforcement Learning

Lecture 2. Probability Basics

Course Instructor: Ling PAN

Department of Electronic and Computer Engineering

Hong Kong University of Science and Technology

Acknowledgment: Most slides are adapted from UC Berkeley CS188 & CS285, Stanford CS109 & CS229

Outline

- Introduction
- Probability Basics

Uncertainty

- The real world is rife with uncertainty!
 - Example: if I leave for HKG 60 minutes before my flight, will I be there in time?
- Problems:
 - Partial observability (road state, other drivers' plans, etc.)
 - Noisy sensors (radio traffic reports, Google maps)
 - Immense complexity of modelling and predicting traffic, security line, etc.
 - Lack of knowledge of world dynamics (will tire burst? will I get in crash?)
- Probabilistic assertions summarize effects of ignorance and laziness
- Combine probability theory + utility theory → decision theory
 - Maximize expected utility: $a^* = \operatorname{argmax}_a \sum_s P(s|a)U(s)$

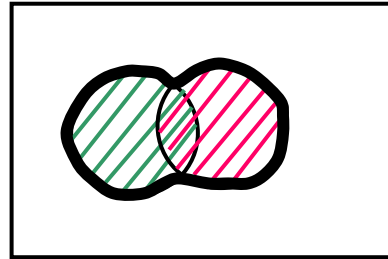
Set Operations

- We use set operations to construct more complex events.

- Union $A \cup B$

" A , B or both occur"

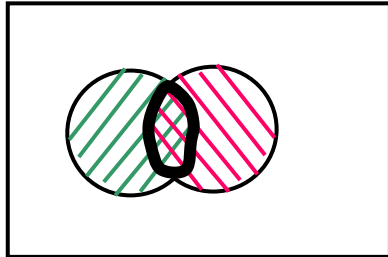
the entire shaded area of both circles



- Intersection $A \cap B$

"Both A and B occur"

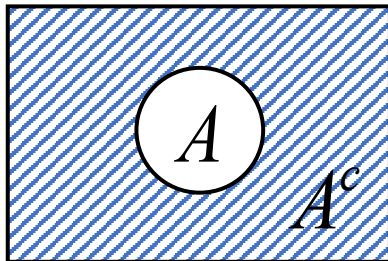
the overlapping area of both circles



- Complement A^c or $\bar{A}$

" A does not occur"

the entire area outside of A



Venn diagram

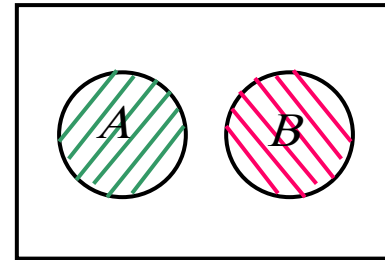
→ Allow us to construct and analyze more complex events

Set Relations

- Equality $A = B$ if they contain the same outcomes

- Mutual Exclusivity $A \cap B = \phi$

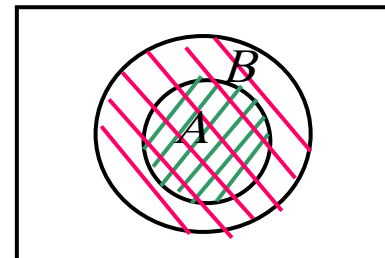
"If A occurs, then B does not"



separate circles with no overlap

- Subset or containment $A \subset B$

"If A occurs, then B occurs"



A as a smaller circle entirely within B



Useful Properties of Set Operations

- Commutative Properties:

$A \cup B = B \cup A$ and $A \cap B = B \cap A$ the order does not matter for union or intersection

- Associative Properties:

$A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$ can group operations however we like without changing the result

- Distributive Properties:

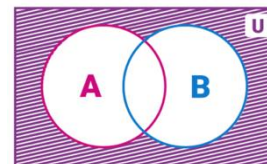
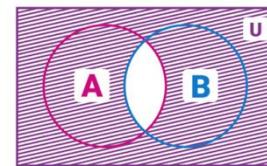
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

allow us to distribute unions over intersections and vice versa

- DeMorgan's Rule:

$(A \cap B)^c = A^c \cup B^c$ and $(A \cup B)^c = A^c \cap B^c$



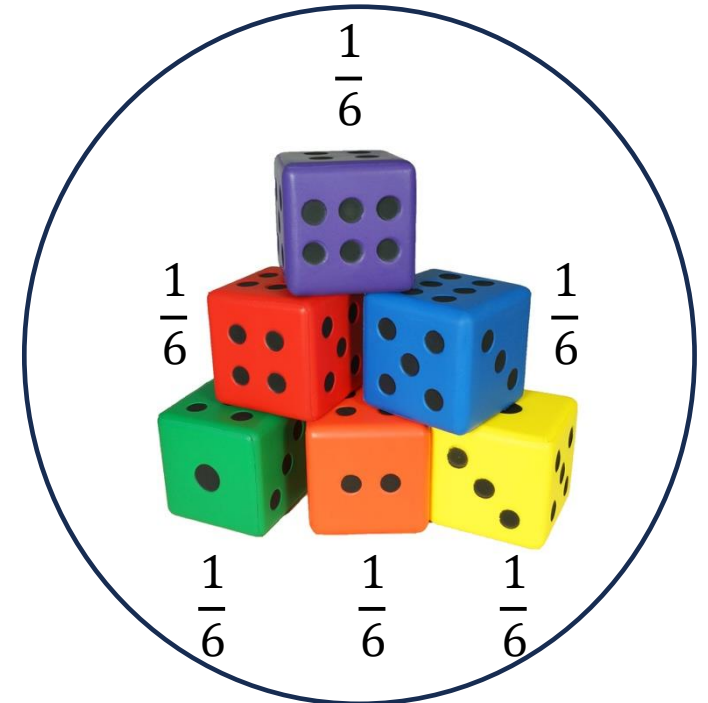
show how complements interact with unions and intersections

Basic Laws of Probability (Discrete)

- Begin with a set Ω of possible outcomes
 - Example: 6 possible rolls of a die $\{1, 2, 3, 4, 5, 6\}$



- A **probability model** assigns a number $P(\omega)$ to each outcome ω
 - Example: $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$
- These numbers must satisfy
 - $0 \leq P(\omega)$
 - $\sum_{\omega \in \Omega} P(\omega) = 1$

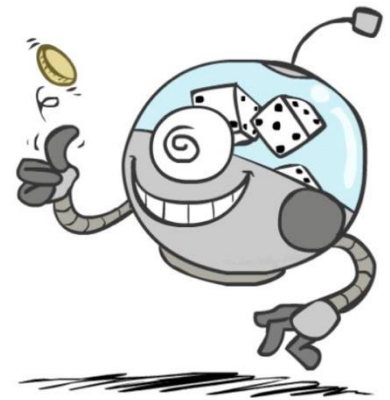


Basic Laws of Probability (Discrete)

- An **event** is any subset of Ω
 - Example: `roll < 4` is the set $\{1, 2, 3\}$
 - Example: `roll is odd` is the set $\{1, 3, 5\}$
- The probability of an event is the **sum** of probabilities over its outcomes

$$P(A) = \sum_{\omega \in A} P(\omega)$$

- Example: $P(\text{roll} < 4) = P(1) + P(2) + P(3) = \frac{1}{2}$



Random Variables

- A random variable is some aspect of the outcomes about which we may be uncertain
- Formally a **deterministic function** of ω (from its **domain** to its **range**)
- The **range** of a random variable is the set of possible values
 - Odd = Is the dice roll an odd number? $\rightarrow \{\text{true}, \text{false}\}$
 - Odd(1)=true, Odd(6) = false
 - Often write the event Odd=true as odd, Odd=false as $\neg\text{odd}$
 - T = Is it hot or cold? $\rightarrow \{\text{hot}, \text{cold}\}$
 - D = How long will it take to get to the airport? $\rightarrow [0, \infty)$
 - L_{Ghost} = Where is the ghost? $\rightarrow \{(0,0), (0,1), \dots\}$
- The **probability distribution** of a random variable X gives the probability for each value x in its range (probability of the event $X = x$)
 - $P(X = x) = \sum_{\{\omega: X(\omega)=x\}} P(\omega)$
 - $P(x)$ for short (when unambiguous)
 - $P(X)$ refers to the entire distribution (think of it as a vector or table)

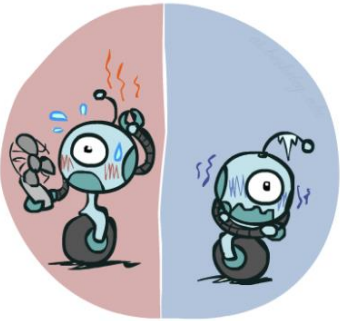
Probability Distributions

- Associate a probability with each value; sums to 1

- Temperature T :

T	P
hot	0.5
cold	0.5

$P(T)$



- Weather W :

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

$P(W)$



← Marginal
distributions

- Joint distribution

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

$P(T, W)$

- Can't deduce joint from marginals
- Can deduce marginals from joint

Probabilities of Events

- Recall that the probability of an event is the sum of probabilities of its outcomes:

$$P(A) = \sum_{\omega \in A} P(\omega)$$

→ Given a joint distribution over all variables, can compute any event probability:

- Probability that it's hot AND sunny?
- Probability that it's hot?
- Probability that it's hot OR not foggy?

- Joint distribution

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

$P(T, W)$

Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): collapse a dimension by adding

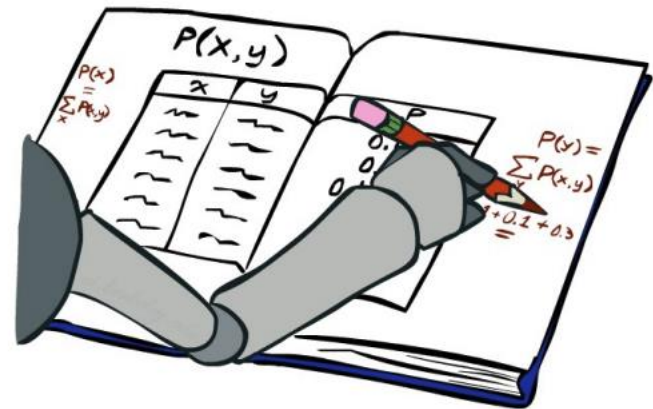
$$P(X = x) = \sum_y P(X = x, Y = y)$$

- Joint distribution $P(T, W)$

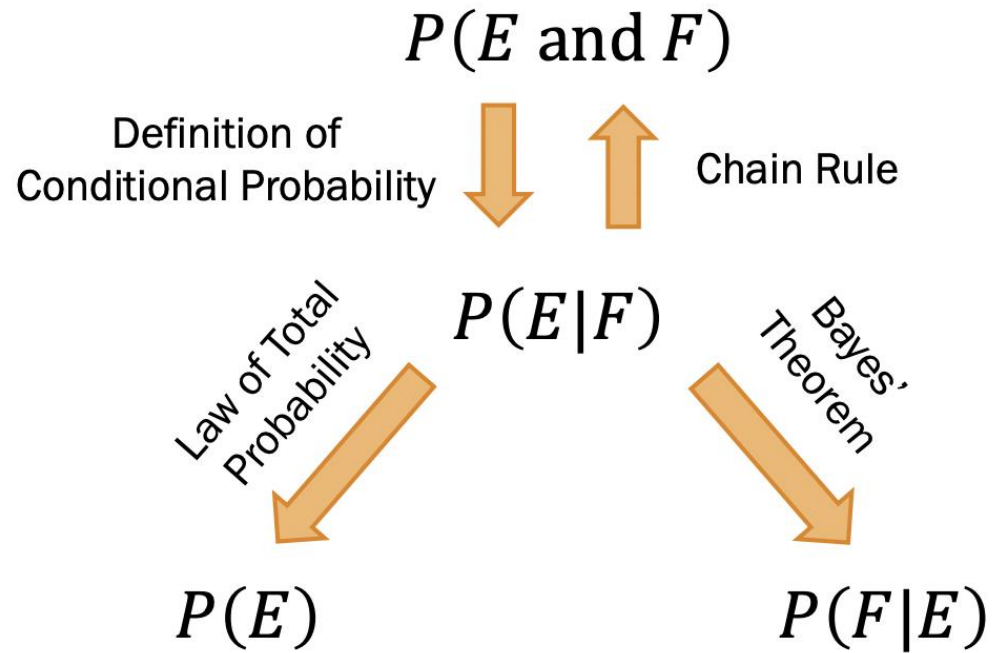
		Temperature		
		hot	cold	
Weather	sun	0.45	0.15	0.6
	rain	0.02	0.08	0.1
	fog	0.03	0.27	0.3
	meteor	0.00	0.00	0.0
		0.5	0.5	

$P(W)$

$P(T)$



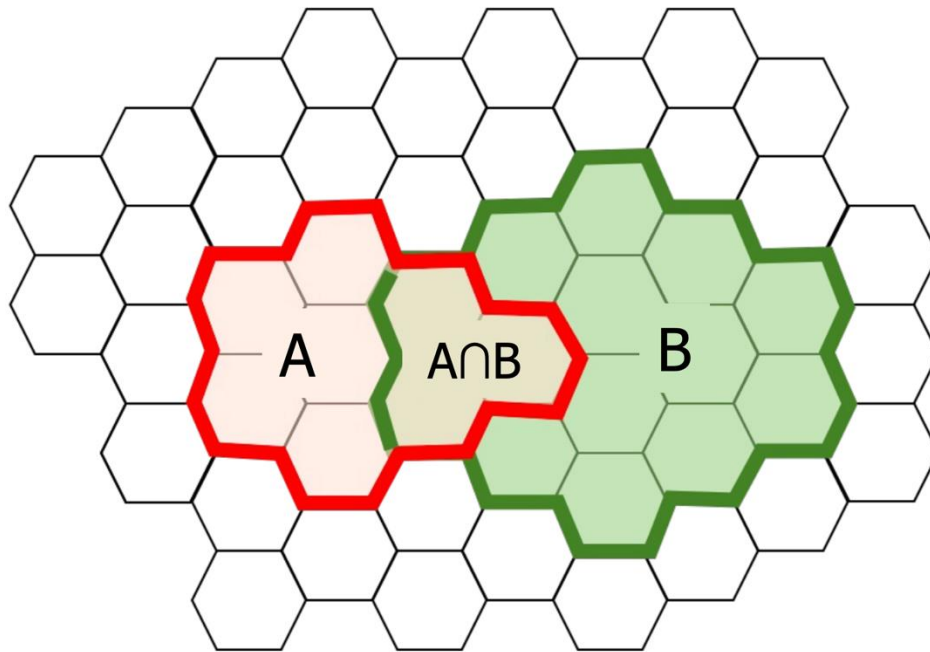
Conditional Probabilities



Conditional Probabilities

- The **conditional probability** of event A given an event B is the probability that A occurs, given that B occurs. This is called “conditioning on B ”
 - A simple relation between joint and conditional probabilities

The probability of
A given B $P(A|B) = \frac{P(A, B)}{P(B)}$, where we assume that $P[B] > 0$



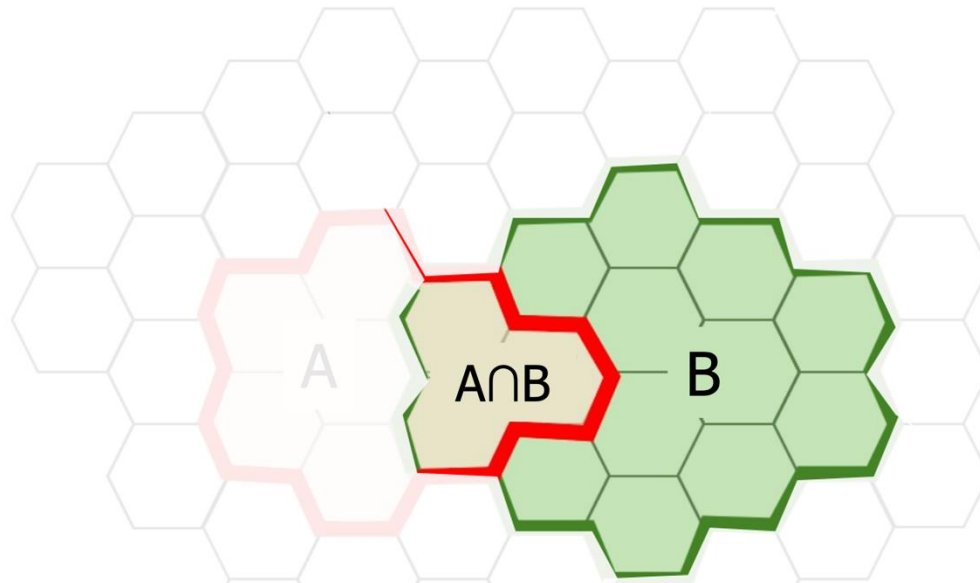
$$P(A) = \frac{8}{50} = 0.16$$

$$P(A|B) = ?$$

Conditional Probabilities

- The **conditional probability** of event A given an event B is the probability that A occurs, given that B occurs. This is called “conditioning on B ”
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The probability of
A given B $P(A|B) = \frac{P(A, B)}{P(B)}$, where we assume that $P[B] > 0$



$$P(A) = \frac{8}{50} = 0.16$$

$$P(A|B) = \frac{3}{14} \approx 0.21$$

$$P(A|B) = \frac{\text{\#of outcomes in } A \text{ consistent with } B}{\text{\#of outcomes in } S \text{ consistent with } B} = \frac{|A \cap B|}{|S \cap B|} = \frac{|A \cap B|}{|B|}$$

Conditional Probabilities

- Example:

$P(T, W)$

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

$$\begin{aligned} P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\ &= \frac{P(W = s, T = c)}{\sum_w P(W = w, T = c)} \\ &= \frac{0.15}{0.50} = 0.3 \end{aligned}$$

Normalizing a Distribution

- To bring or restore to a normal condition (all entries sum to 1)
- Procedure
 - Multiply each entry by $\alpha = \frac{1}{\text{sum over all entries}}$

$P(W, T)$

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

$$P(W, T = c) \quad P(W|T = c) = \frac{P(W, T = c)}{P(T = c)} = \alpha P(W, T = c)$$

0.15
0.08
0.27
0.00

Normalize
 $\xrightarrow{\alpha = \frac{1}{0.5} = 2}$

0.30
0.16
0.54
0.00

Conditional Distributions

- Distributions for one set of variables given another set

$$P(T, W)$$

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

$$P(W|T = h)$$

hot

0.90
0.04
0.06
0.00

$$P(W|T = c)$$

cold

0.30
0.16
0.54
0.00

$$P(W|T)$$

hot	cold
0.90	0.30
0.04	0.16
0.06	0.54
0.00	0.00

The Product (or Chain) Rule

- Sometimes have conditional distributions but want the joint

$$\begin{array}{llll} P(A|B) = \frac{P(A, B)}{P(B)} & \xrightarrow{\text{Multiply both sides by } P(B)} & P(A|B)P(B) = P(A, B) \\ P(B|A) = \frac{P(A, B)}{P(A)} & \xrightarrow{\text{Multiply both sides by } P(A)} & P(B|A)P(A) = P(A, B) \end{array}$$

 switch B and A

- ✓ Can compute the joint probability using either conditional probability, depending on which one we know or can more easily calculate.
 - ✓ Allows us to break down complex joint probabilities into more manageable pieces (It's often easier to measure conditional probabilities and individual event probabilities than it is to directly measure joint probabilities).
- Bayesian inference: update our beliefs based on new evidence

The Product (or Chain) Rule

- Sometimes have conditional distributions but want the joint

$$P(A|B)P(B) = P(A, B) \xleftarrow[\text{sides by } P(B)]{\text{Multiply both}} P(A|B) = \frac{P(A, B)}{P(B)}$$

- Example: $P(W|T)P(T) = P(W, T)$

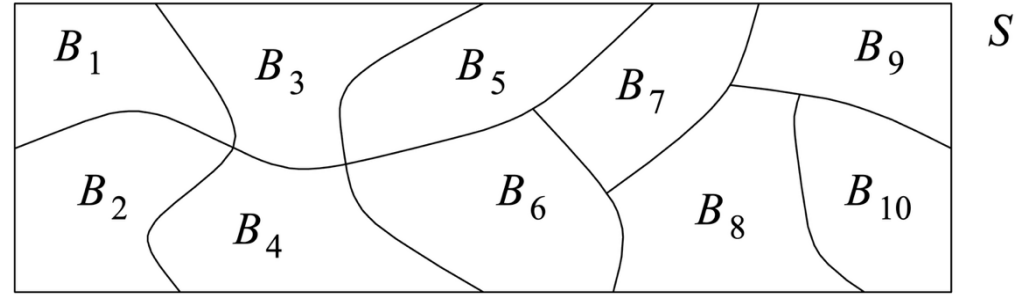
$P(W T)$		$P(T)$		$P(T, W)$			
0.90	0.30	T	P			Temperature	
0.04	0.16	hot	0.5			hot	cold
0.06	0.54	cold	0.5	Weather		0.45	0.15
0.00	0.00					0.02	0.08
						0.03	0.27
						0.00	0.00
hot	cold						

The Product (or Chain) Rule

- A joint distribution can be written as a product of conditional distributions by repeated application of the product rule
 - $P(X_1, X_2, X_3) = P(X_3|X_1, X_2)P(X_1, X_2) = P(X_3|X_1, X_2)P(X_2|X_1)P(X_1)$
 - $P(X_1, X_2, \dots, X_n) = \prod_i P(X_i|X_1, \dots, X_{i-1})$

Total Probability Theorem

- Partitioning the sample space:
 - Definition: $B_1, \dots, B_n$ form a partition of the sample space S if both of the following are true
 - $B_1, \dots, B_n$ are mutually exclusive
 - Their union equals S , i.e., $\bigcup_{i=1}^n B_i = S$



- If $B_1, B_2, \dots, B_n$ partition the sample space, then for any event A ,

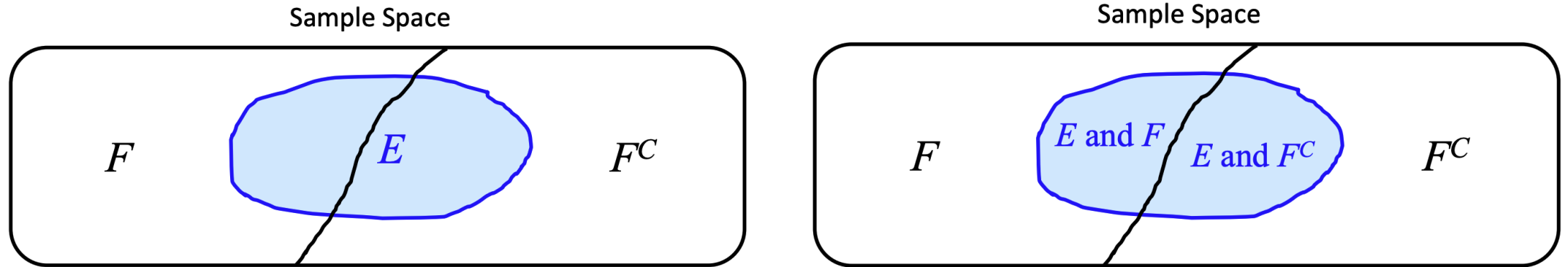
$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$$

Total Probability Theorem

- If $B_1, B_2, \dots, B_n$ partition the sample space, then for any event A ,

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$$

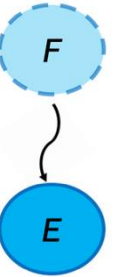
- What does it mean in the simplest case when $n = 2$?



$$\begin{aligned} P(E) &= P(EF) + P(EF^C) \\ &= P(E|F)P(F) + P(E|F^C)P(F^C) \end{aligned}$$

Guide to notion: "And" $P(E \text{ and } F), P(E \cap F), P(E, F), P(EF)$

Bayes' Rule $P(E|F) \rightarrow P(F|E)$



- I want to calculate $P(\text{State of the world } F \mid \text{Observation } E)$
 - It seems so tricky...
 - Requires us to reason backwards from effects to causes
 - If we observe symptoms (E), determining the exact probability of a specific disease (F) can be complex, as multiple diseases might cause similar symptoms
- The other way around is easy $P(\text{Observation } E \mid \text{State of the world } F)$
 - Follows the natural flow of cause and effect
 - Easier for doctors to know the probability of certain symptoms appearing (E) given a specific disease (F)

Bayes' Rule $P(E|F) \rightarrow P(F|E)$



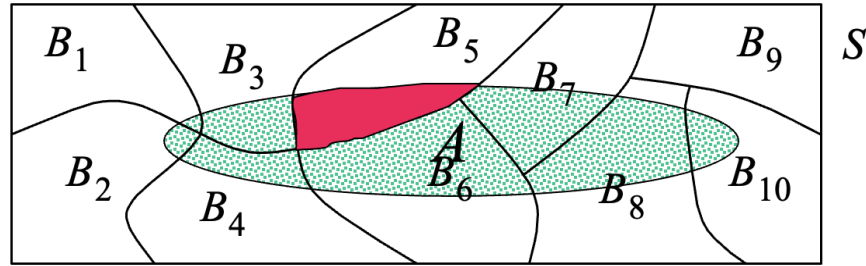
$$P(F|E) = \frac{P(EF)}{P(E)} \quad \text{Definition of conditional probability}$$

$$= \frac{P(E|F)P(F)}{P(E)} \quad \text{The product (or chain) rule is symmetric}$$

$$= \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|F^C)P(F^C)} \quad \text{Law of total probability}$$

Bayes' Rule $P(E|F) \rightarrow P(F|E)$

- Let $B_1, B_2, \dots, B_n$ be a partition of the sample space S .



- For any event A ,

$$P(B_j|A) = \frac{P(AB_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{k=1}^n P(A|B_k)P(B_k)}$$

- Bayes' rule lets us **switch** which event is the conditioning event.
- This simplifies the calculation when the $P[A|B_k]$ are easy to compute.

Independence

- Two events A and B are independent if $P(AB) = P(A) \times P(B)$
 - Meaning: the probability of their joint occurrence is equal to the product of their individual probabilities
- Equivalently, an event A is independent of event B if knowledge that B occurs does not alter the probability of event A and vice versa:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A) \quad P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A)P(B)}{P(A)} = P(B)$$

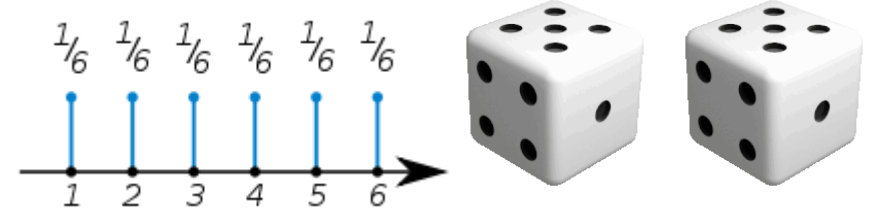
→ The conditional probability of A given B is just the probability of A .

→ The conditional probability of B given A is just the probability of B .

Independence

- Example: two dice rolls Roll_1 and Roll_2

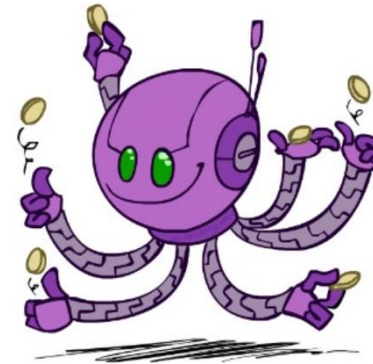
- $P(\text{Roll}_1 = 5, \text{Roll}_2) = P(\text{Roll}_1 = 5)P(\text{Roll}_2 = 3) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$
 - $P(\text{Roll}_2 = 3 | \text{Roll}_1 = 5) = P(\text{Roll}_2 = 3)$



- Example: n fair, independent coin flips:

$P(X_1)$			$P(X_n)$	
H	T		H	T
0.5	0.5		0.5	0.5

$\underbrace{\hspace{15em}}_{P(X_1, \dots, X_n)}$



- Independence is incredibly powerful
 - Exponential reduction in representation size
- Independence is extremely rare
- Conditional independence is ubiquitous

Conditional Independence

- Conditional independence is our most basic and robust form of knowledge about uncertain environments.
- X is conditionally independent of Y given Z if and only if:

$$P(X|Y, Z) = P(X|Z), \quad \forall X, Y, Z$$

or equivalently, if and only if

$$P(X, Y|Z) = P(X|Z)P(Y|Z), \quad \forall X, Y, Z$$

Expectation

- In order to completely describe the behavior of a discrete random variable, an entire probability mass function (PMF) $p_X(x)$ must be given.
- In some situations we are interested in a few parameters that summarize the information provided by the PMF.
- What is the average value we could expect some random variable to be?

A possible value of X

The probability of that value

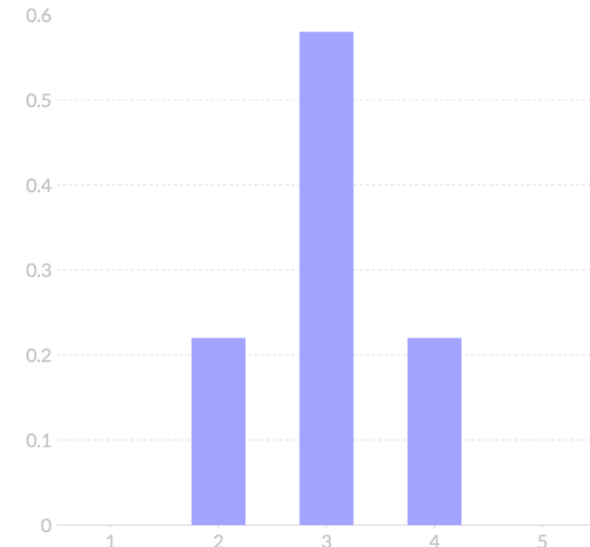
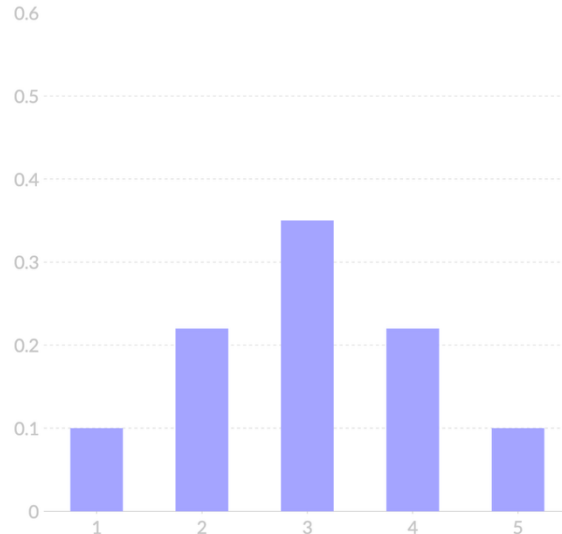
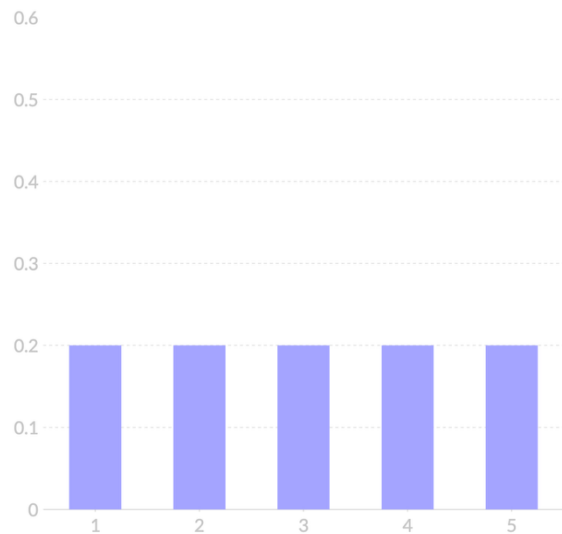
$$m_X = E[X] = \sum_{x \in S_X} xp_X(x) = \sum_k x_k p_X(x_k)$$

Loop over all values x that X can take on

The diagram shows the formula for the expectation of a discrete random variable X . The formula is enclosed in a light blue rounded rectangle. Above the rectangle, two arrows point to the terms x and $p_X(x)$ in the first sum, with labels 'A possible value of X ' and 'The probability of that value' respectively. Below the rectangle, an arrow points to the summation index $x \in S_X$ with the label 'Loop over all values x that X can take on'.

Intuition: Measure of Spread

- Expectation is not all you need
- Consider the following 3 distributions (PMFs)
 - All have the same expected value $\mathbb{E}[X]$
 - But “spread” in distribution is different



- The variance of a random variable measures how much the values of the random variable generally differ from the mean.

Variance

- Define $D = X - \mathbb{E}[X]$ to be the difference between the value of the random variable and its mean (note that this can be positive or negative)
- The variance of a random variable, $\text{VAR}[X]$, is defined as the expected value of the squared difference:

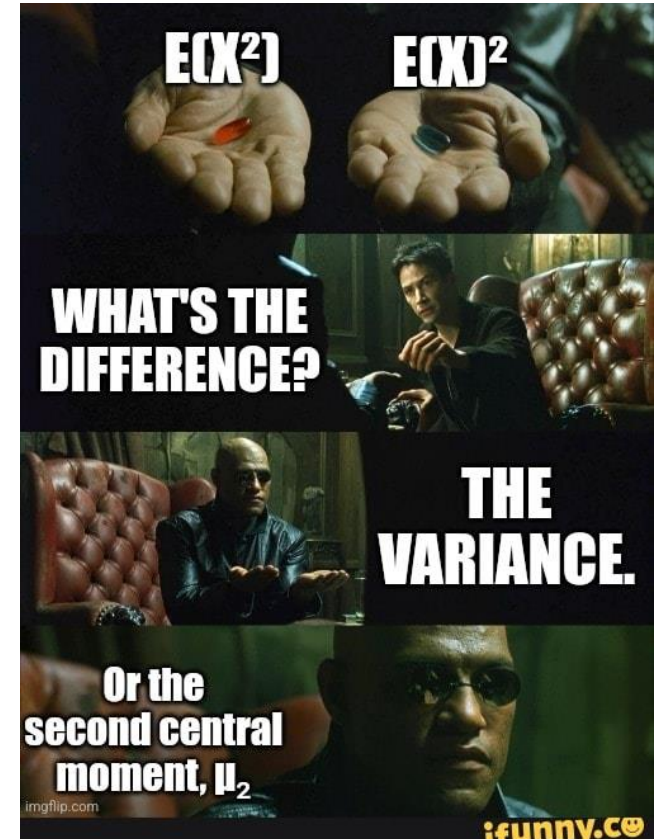
How far away from the mean is X , on average?

On average The random variable X The mean of X

$$\text{VAR}[X] = E[D^2] = E[(X - E[X])^2]$$

distance

- Property: $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$



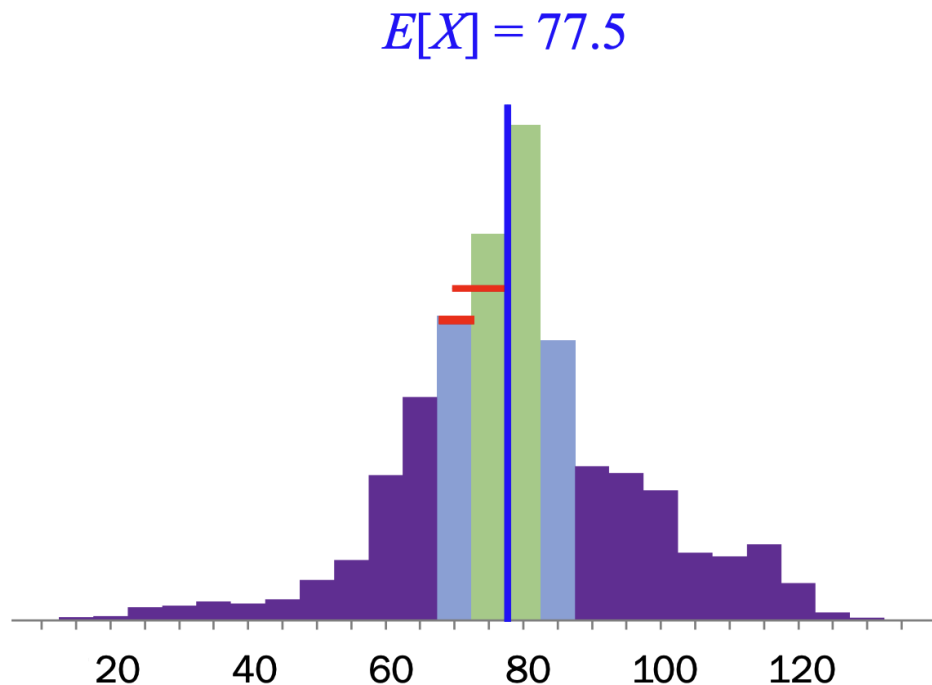
Variance Intuition

- Let X be a random variable that represents a midterm exam grade.

How far away from the mean is X , on average?

On average The RV X The mean of X

$$\text{VAR}[X] = E[D^2] = E[\underbrace{(X - E[X])^2}_{\text{distance}}]$$



X	$(X - \mu)^2$
45 points	1056 points ²
100 points	506 points ²
70 points	56 points ²
...	

$$\text{Var}(X) = 52 \text{ points}^2$$

Standard Deviation

- The standard deviation of a random variable, $\text{STD}[X]$, is defined as the square root of the variance
 - The advantage of the standard deviation is that its units are the same as that of X .
 - For example, the variance of height would be in units of meters^2 , but the standard deviation would be in units of meters
 - More intuitive to understand in the context of the original data

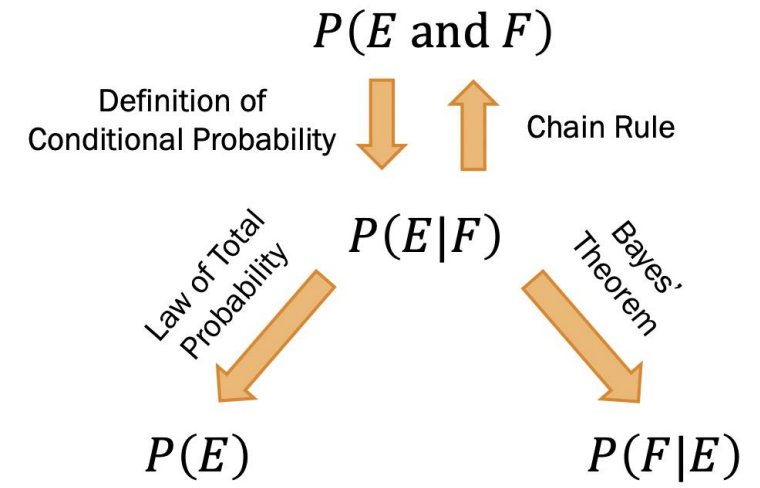
$$\text{STD}[X] = \sqrt{\text{VAR}[X]}$$

Units are the same as
your random variable

Units are squared



The Core Probability Toolkit ✓



Definition of Conditional Probability

$$P(E|F) = \frac{P(E \text{ and } F)}{P(F)}$$

The Law of Total Probability

$$P(E) = P(E \text{ and } F) + P(E \text{ and } F^C)$$

$$P(E) = \sum_{i=1}^n P(E \text{ and } B_i)$$

$$P(E) = P(E|F)P(F) + P(E|F^C)P(F^C)$$

$$= \sum_{i=1}^n P(E|B_i)P(B_i)$$

Bayes' Theorem

$$P(B|E) = \frac{P(E|B) \cdot P(B)}{P(E)}$$

$$P(B|E) = \frac{P(E|B) \cdot P(B)}{P(E|B) \cdot P(B) + P(E|B^C) \cdot P(B^C)}$$

Independence

$$P(E|F) = P(E)$$

$$P(E \text{ and } F) = P(E)P(F)$$

Axiom 1: $0 \leq P(E) \leq 1$

Axiom 2: $P(S) = 1$

Axiom 3: If E and F are mutually exclusive, then $P(E \text{ or } F) = P(E) + P(F)$

Otherwise, use Inclusion-Exclusion:

$$P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$$

$$P(E^C) = 1 - P(E)$$

Chain Rule

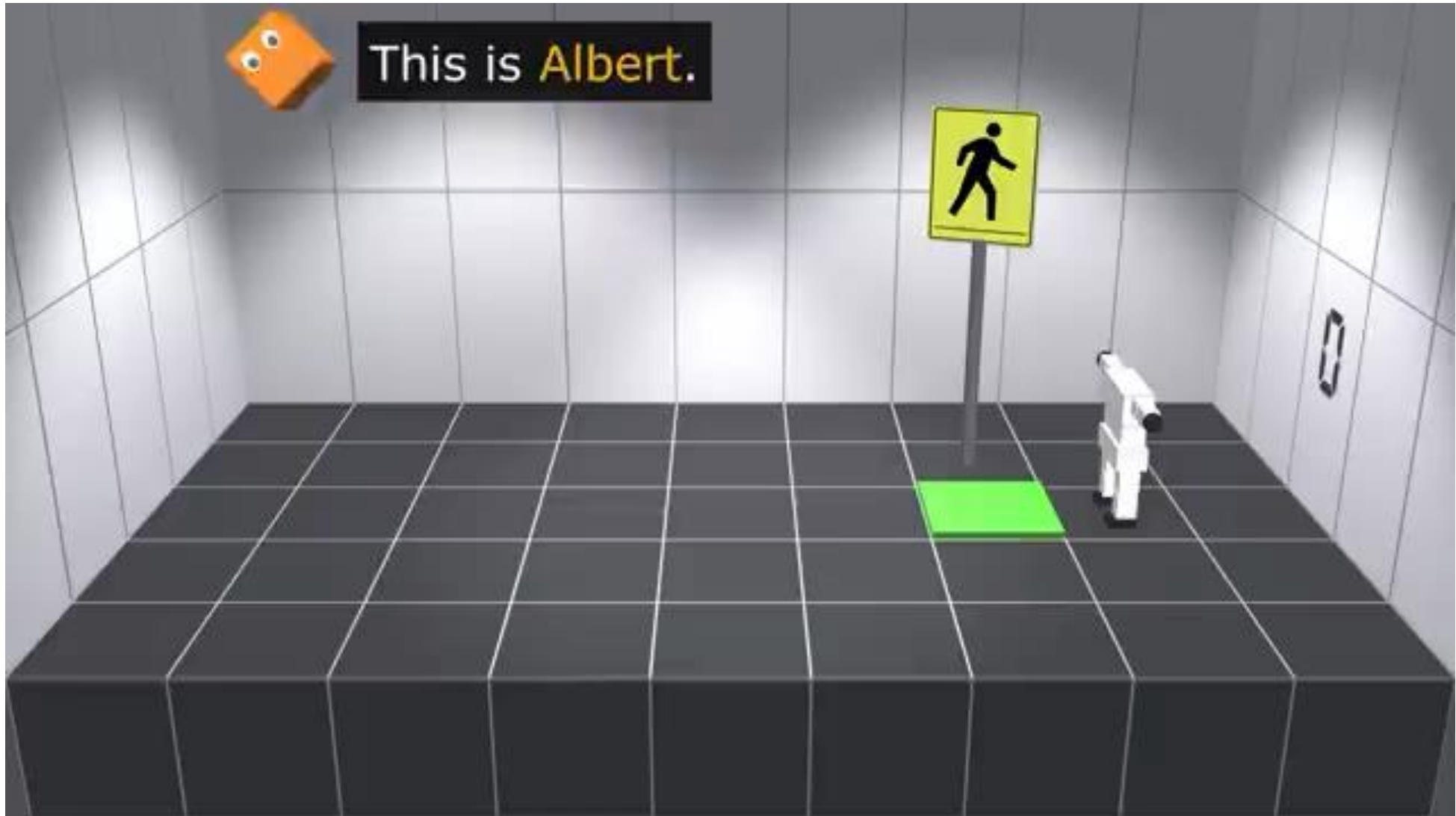
$$\begin{aligned} P(E \text{ and } F) &= P(E|F) \cdot P(F) \\ &= P(F|E) \cdot P(E) \end{aligned}$$

De Morgan's Laws

$$(A \text{ or } B)^C = A^C \text{ and } B^C$$

$$(A \text{ and } B)^C = A^C \text{ or } B^C$$

Learning Challenges



Thanks for your listening.